

2016.0 RANGE ROVER (LG), 502-00

UNI-BODY, SUBFRAME AND MOUNTING SYSTEM

FRONT SUBFRAME – TDV6 3.0L DIESEL (G1591891)

REMOVAL AND INSTALLATION

SPECIAL TOOL(S)



303-1129

Engine Lifting
Brackets



**JLR-204-
794**

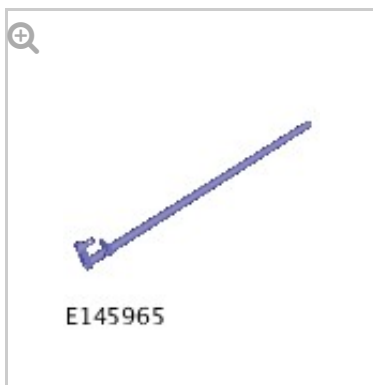
Left Alignment Pin,
Subframe



JLR-204-796

Right Alignment
Pin, Subframe

E145961



JLR-204-802

Splitter, Damper

E145965

REMOVAL

CAUTIONS:

- Make sure that both the Transmission Undertray and Engine Undertray are correctly installed to the vehicle, as they are structural components.
- The final tightening of the suspension components must be carried out with the vehicle on its wheels.

NOTES:

- Some components shown removed for clarity.
- Some variation in the illustrations may occur, but the essential information is always correct.



All vehicles

1.

WARNING:

Make sure to support the vehicle with axle stands.

Raise and support the vehicle.

2.

Disconnect the battery ground cable.

Refer to: [Specifications](#) (414-00 Battery and Charging System - General Information, Specifications).

3.

Refer to: [Radiator](#) (303-03A Engine Cooling - TDV6 3.0L Diesel - Gen 2/TDV6 3.0L Diesel, Removal and Installation).

4.

Refer to: [Generator](#) (414-02A Generator and Regulator - TDV6 3.0L Diesel, Removal and Installation).

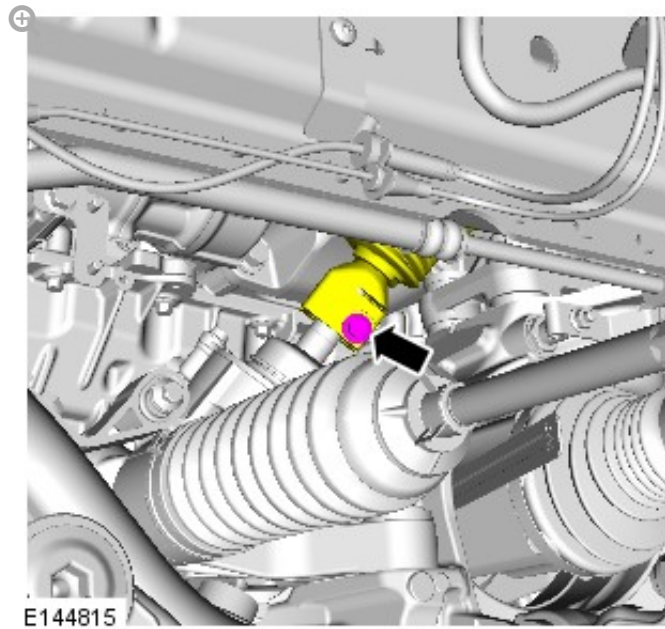
5.

CAUTION:

Discard the bolt.

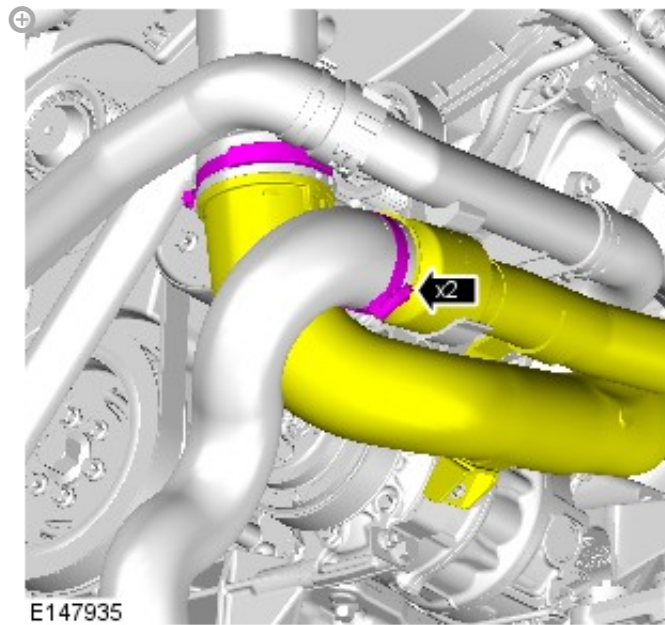
NOTE:

LHD illustration shown, RHD is similar.

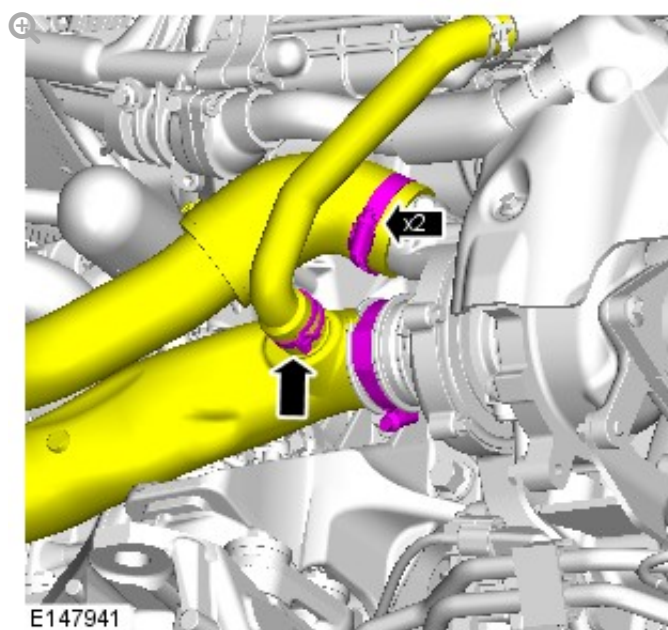


Align the steering wheel to straight ahead.

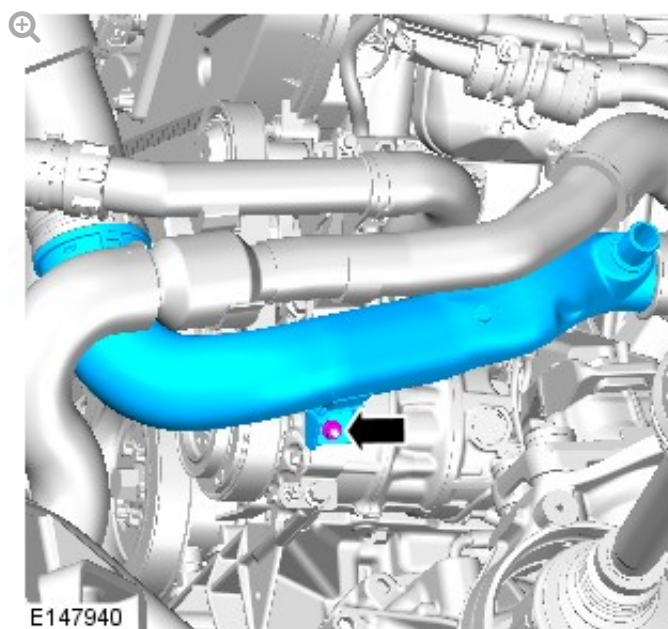
6.



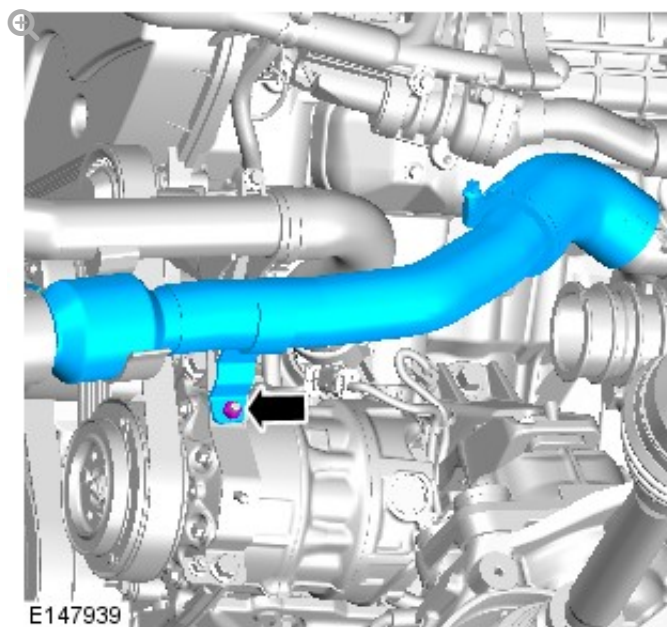
7.



8.



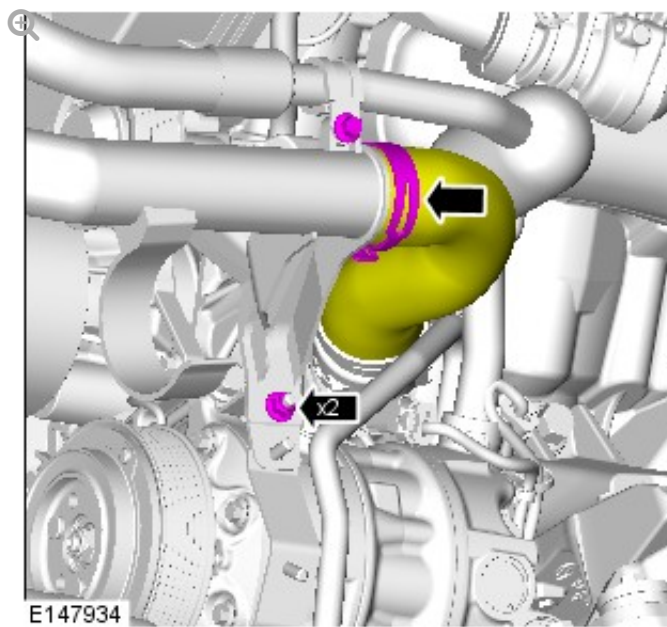
9.



10.

CAUTION:

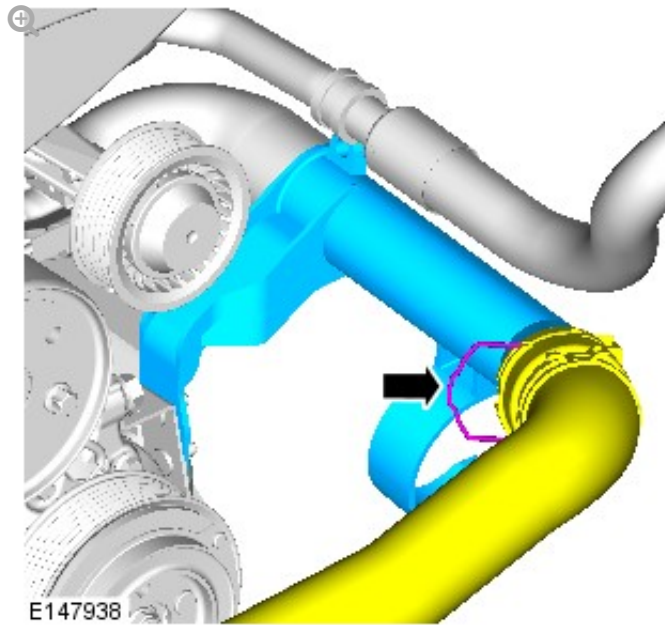
Be prepared to collect escaping coolant.



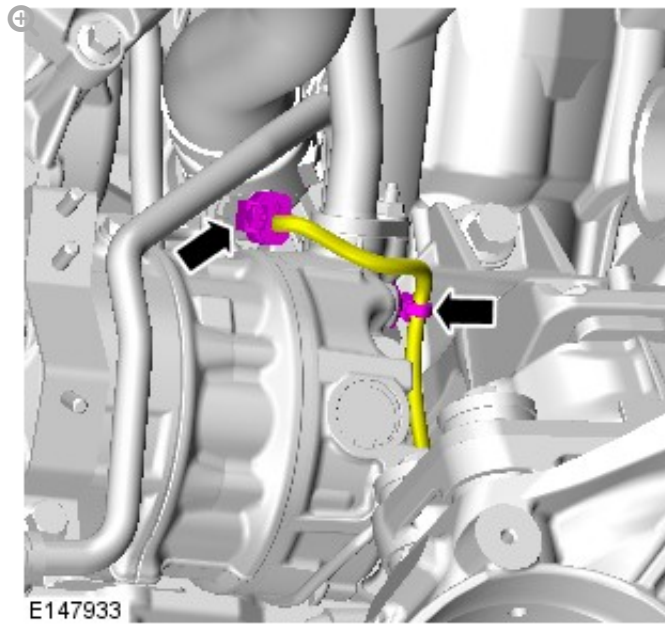
11.

CAUTION:

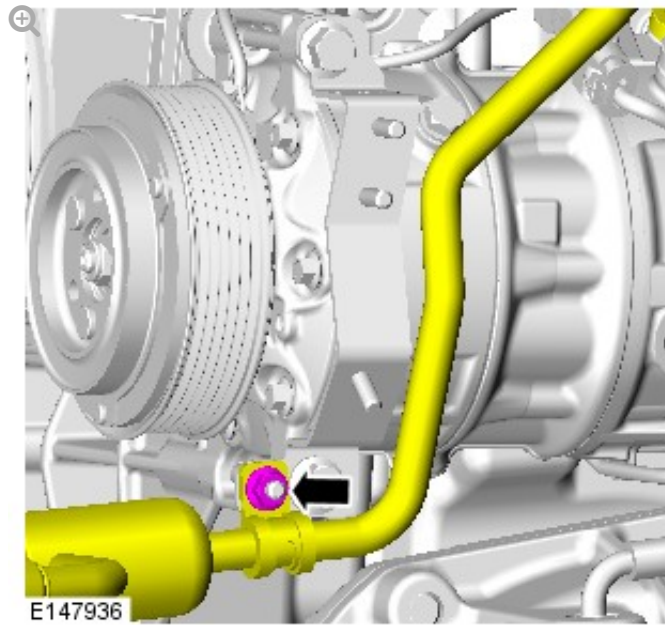
Be prepared to collect escaping coolant.



12.



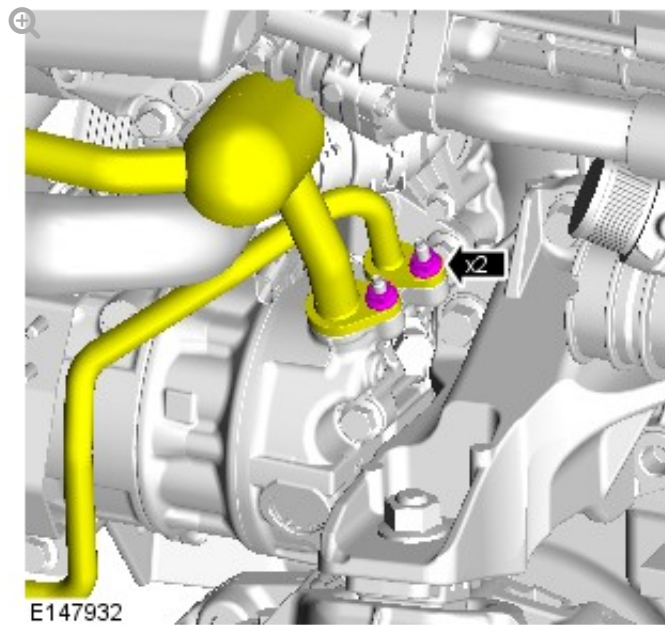
13.



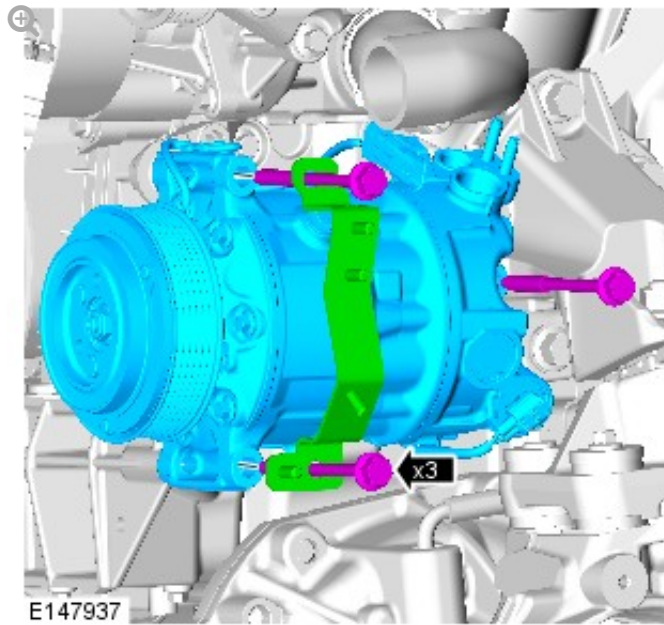
14.

CAUTION:

A new O-ring seal is to be installed.

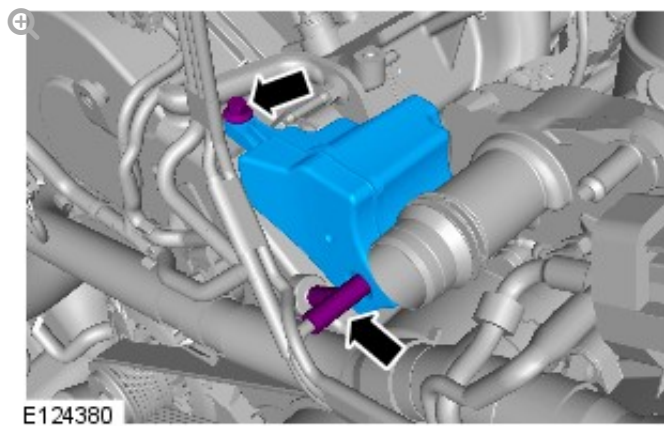


15.

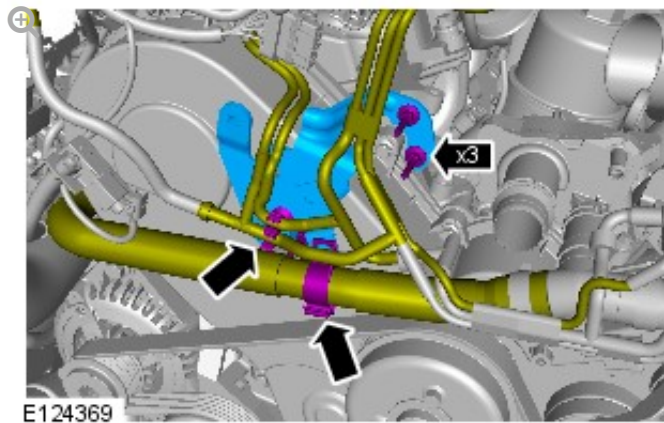


16. Refer to: [Intake Air Shutoff Throttle](#) (303-04A Fuel Charging and Controls - TDV6 3.0L Diesel, Removal and Installation).

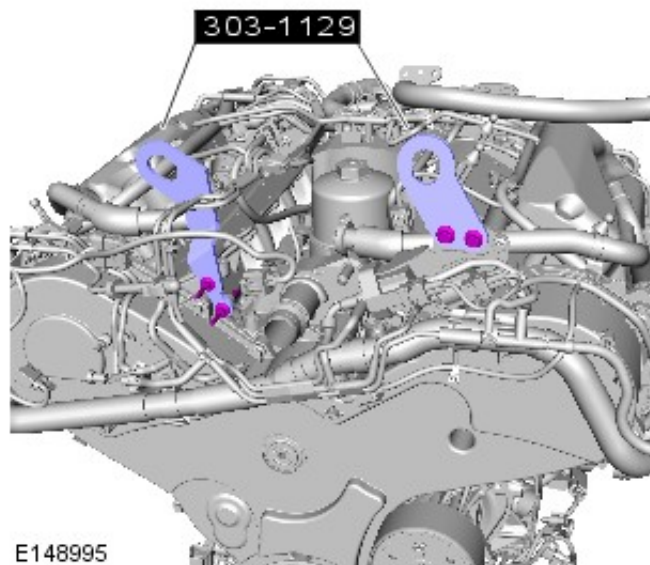
17.



18.



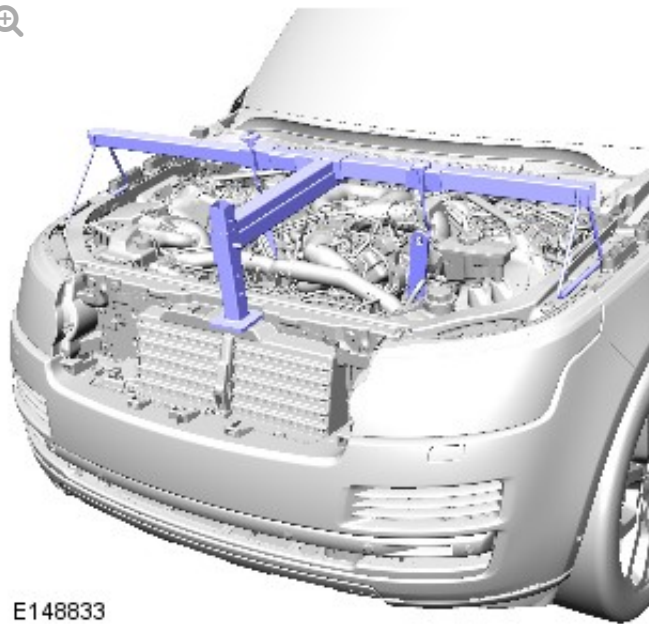
19.



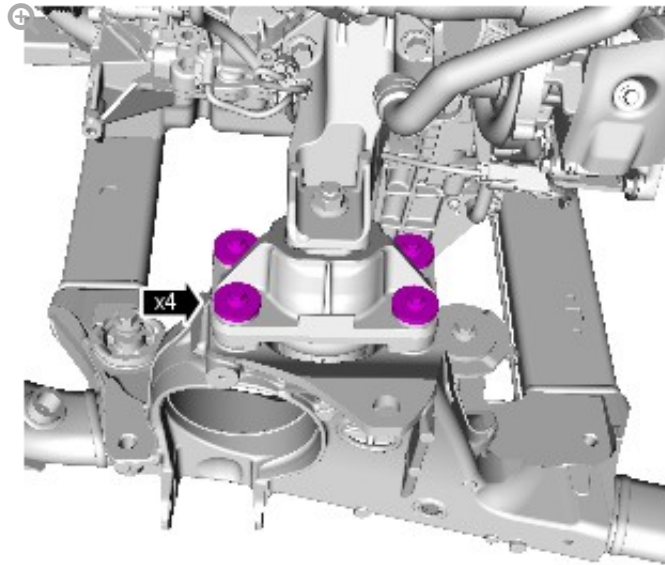
Special Tool(s): [303-1129](#)

Torque: **22 Nm**

20.

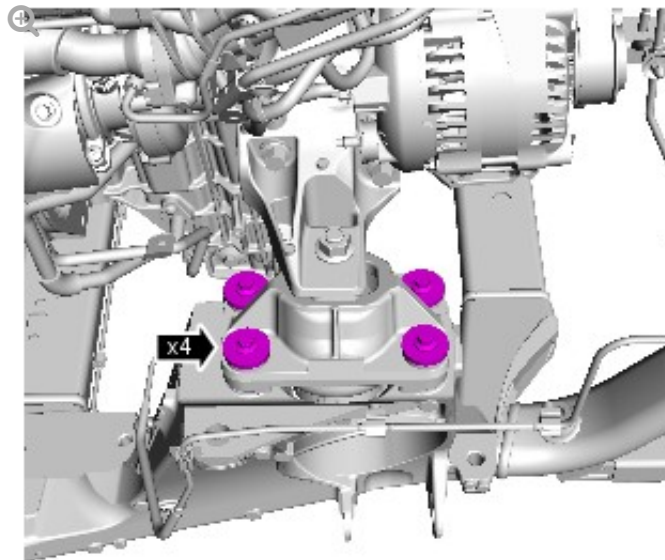


21.



E148990

22.



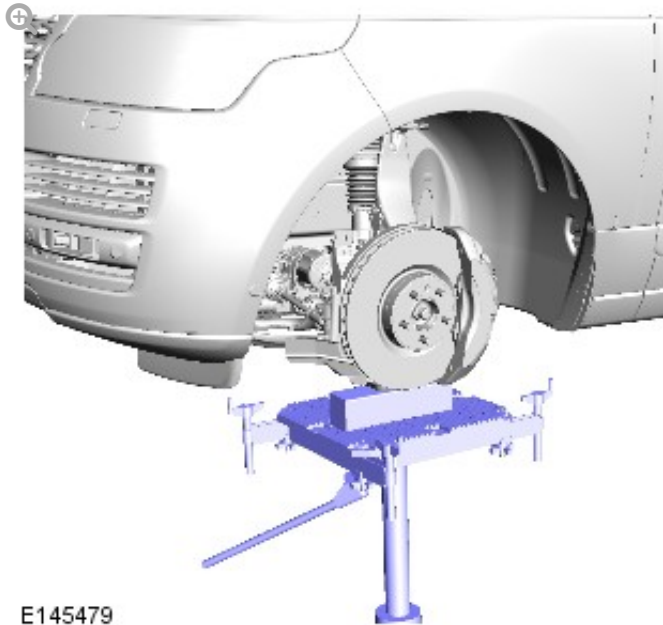
E148992

23. Refer to: [Front Left Halfshaft](#) (205-04 Front Drive Halfshafts, Removal and Installation).
24. Refer to: [Front Right Halfshaft](#) (205-04 Front Drive Halfshafts, Removal and Installation).

25.

NOTE:

LH illustration shown, RH is similar.



26.

WARNING:

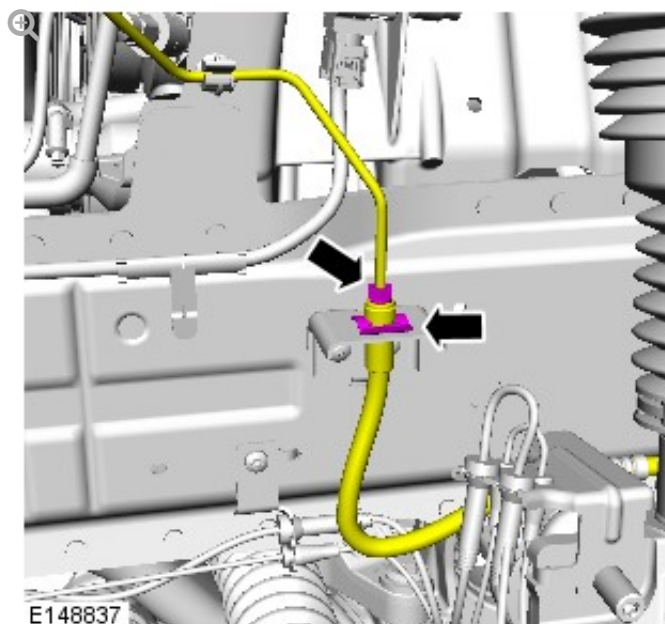
Be prepared to catch escaping fluid.

CAUTION:

Brake fluid will damage paint finished surfaces. If spilled, immediately remove the fluid and clean the area with water.

NOTE:

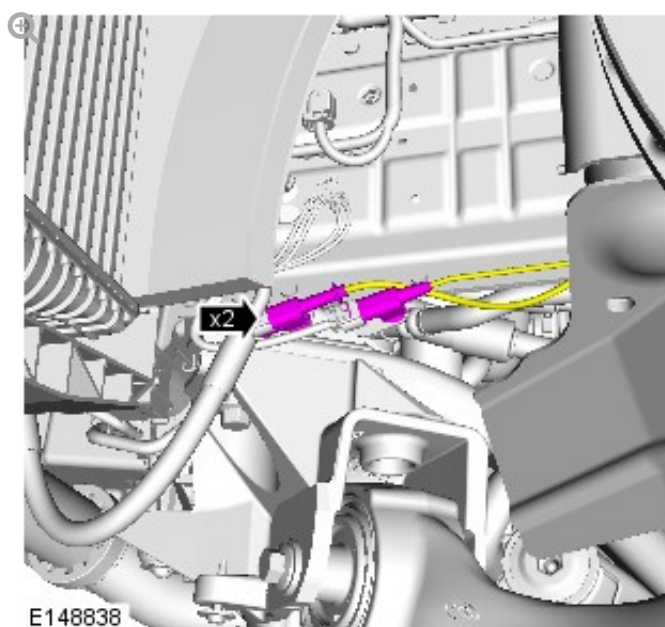
LH illustration shown, RH is similar.



27.

NOTE:

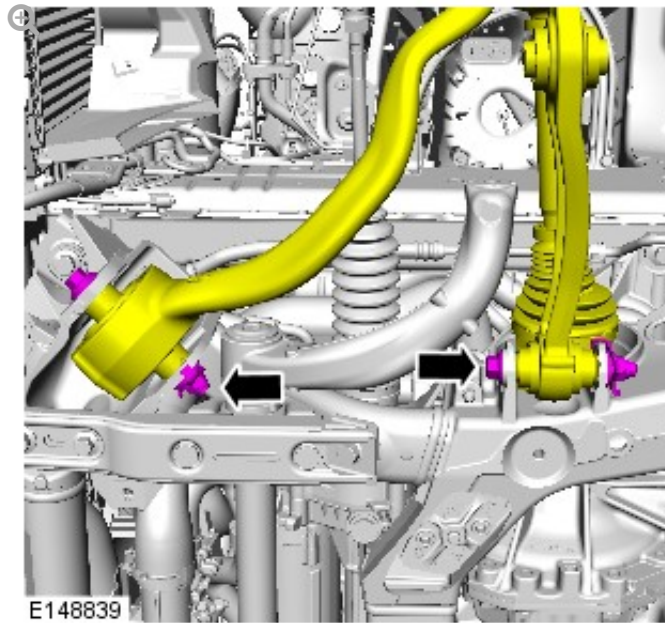
LH illustration shown, RH is similar.



28.

NOTES:

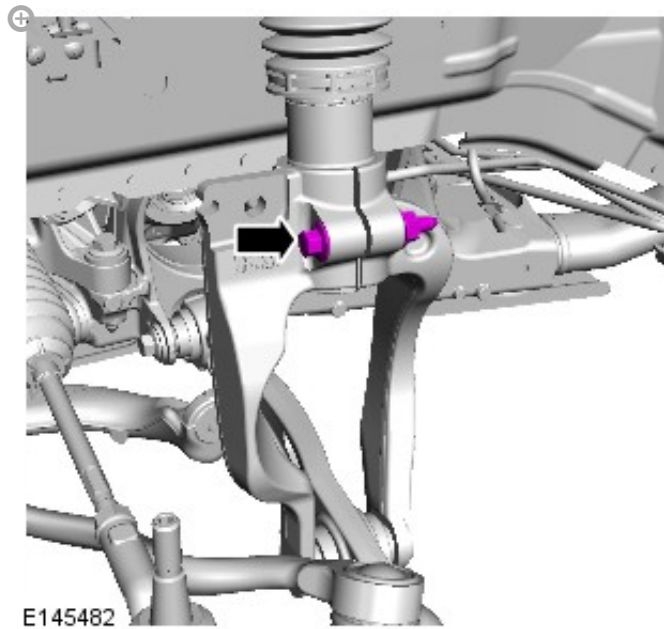
- Discard the nuts and bolts.
- LH illustration shown, RH is similar.



29.

NOTE:

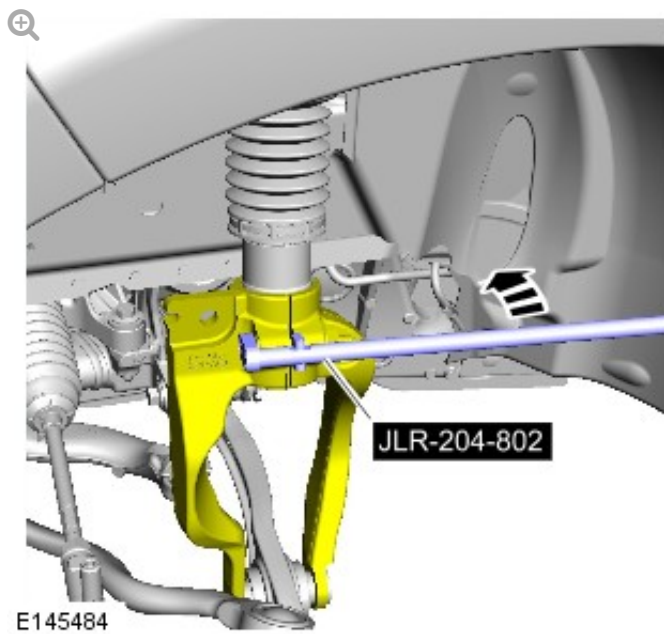
LH illustration shown, RH is similar.



30.

NOTE:

LH illustration shown, RH is similar.

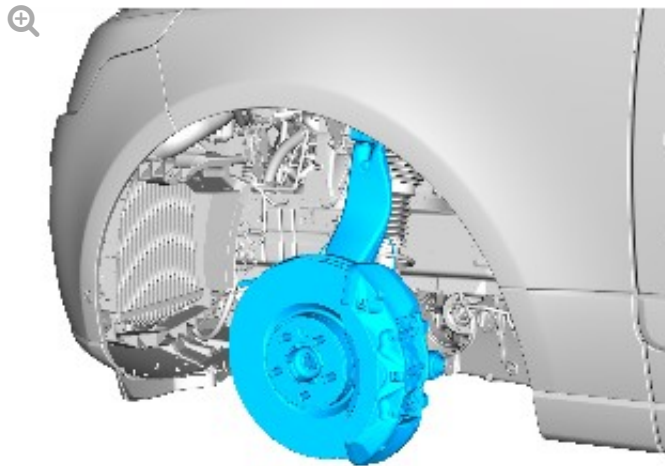


Special Tool(s): [JLR-204-802](#)

31.

NOTE:

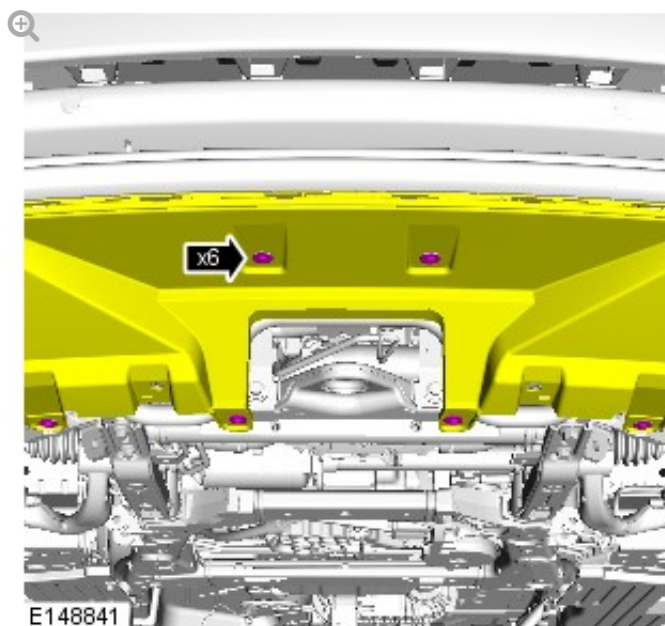
LH illustration shown, RH is similar.



E148840

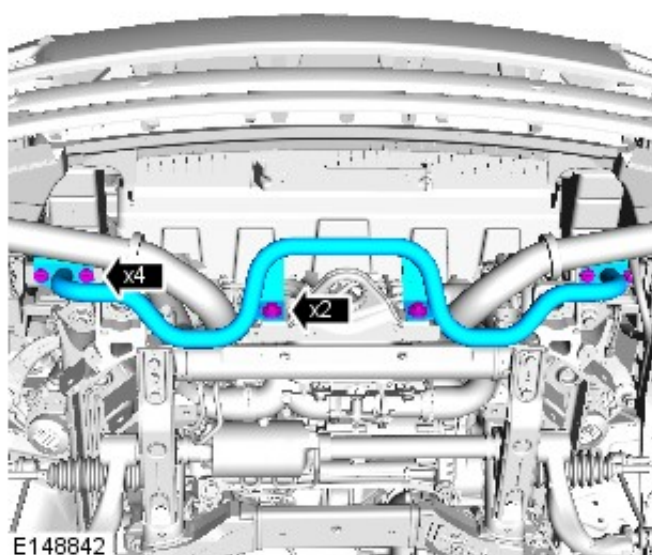
32. Repeat the above 7 steps for the other side.

33.

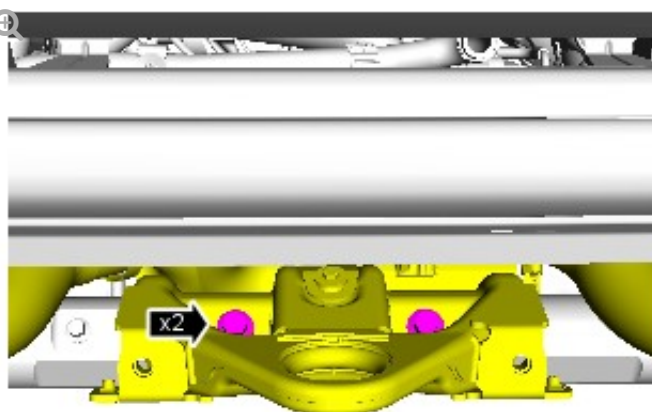


E148841

34.

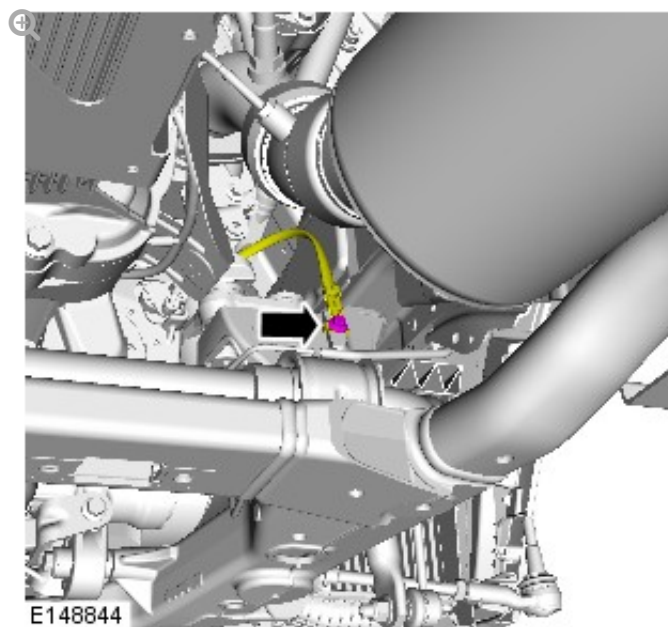


35.

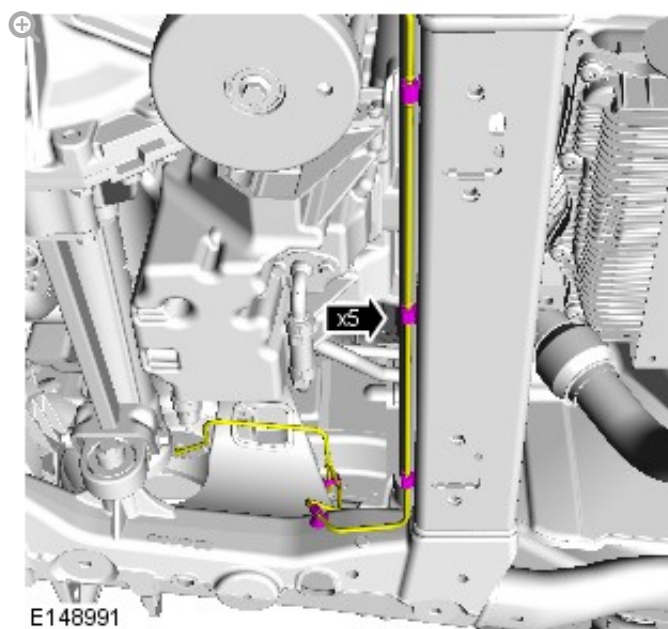


E148843

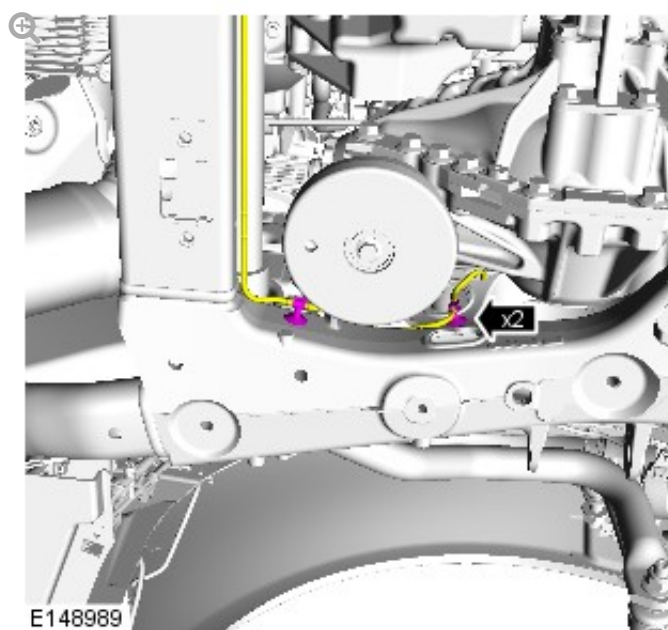
36.



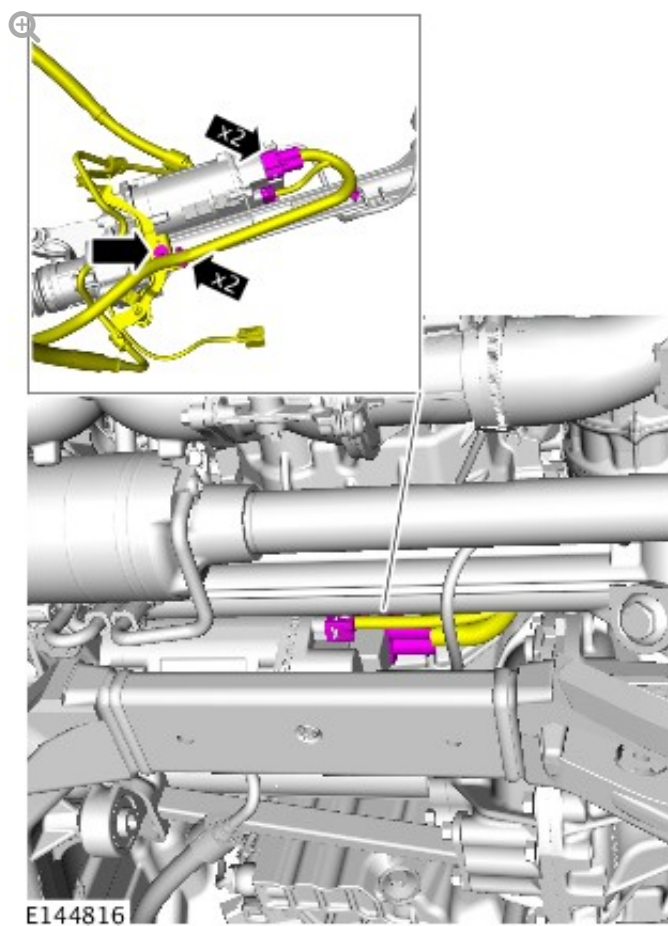
37.



38.



39.



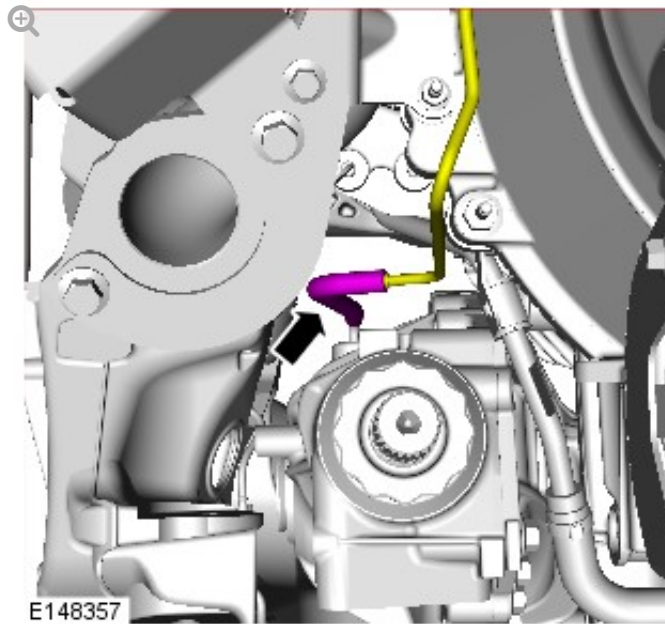
40.

CAUTION:

The transmission must be kept supported while the transmission support crossmember is removed.

Refer to: [Front Driveshaft - TDV6 3.0L Diesel](#) (205-01 Driveshaft, Removal and Installation).

41.



42.

NOTE:

Repeat the procedure for the other side.

Refer to: [Rear Door Lower Moulding - Standard Wheel Base](#) (501-08 Exterior Trim and Ornamentation, Removal and Installation).

Refer to: [Rear Rocker Panel Moulding - Long Wheel Base](#) (501-08 Exterior Trim and Ornamentation, Removal and Installation).



Short wheelbase

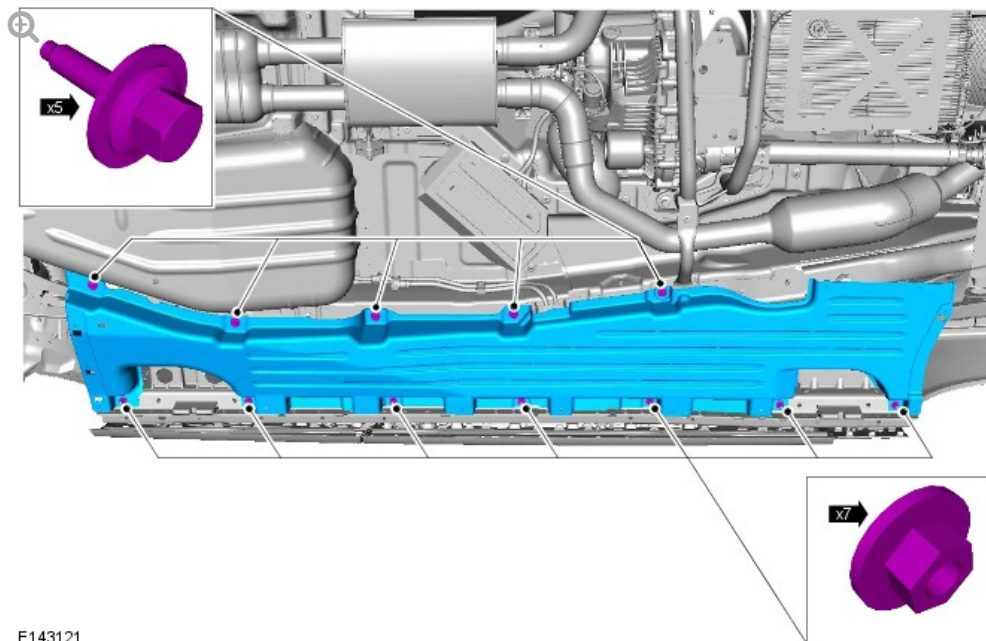
43.

CAUTION:

LH illustration shown, RH is similar.

NOTE:

Repeat the procedure for the other side.



Long wheelbase

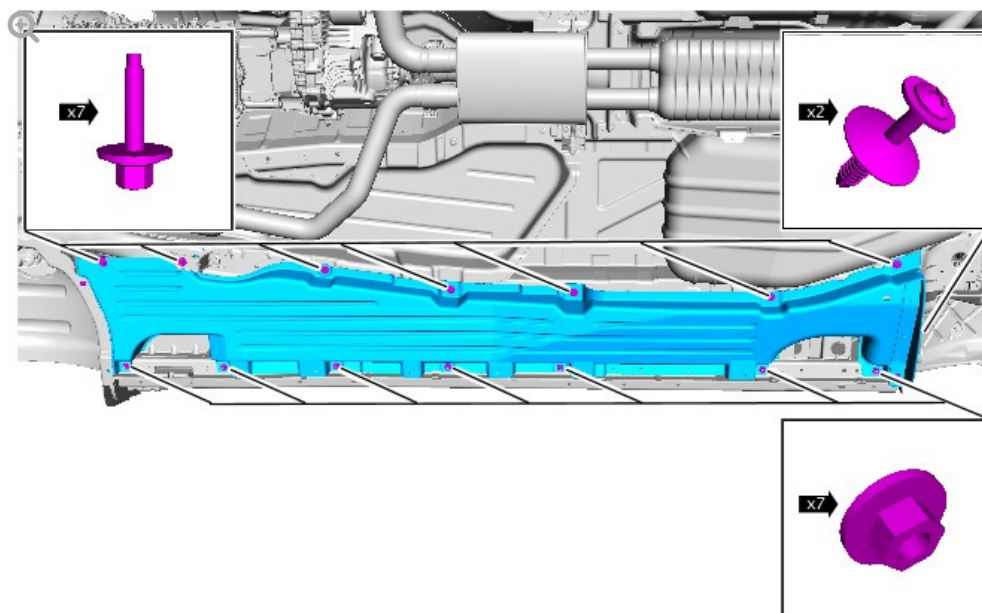
44.

CAUTION:

RH illustration shown, LH is similar.

NOTE:

Repeat the procedure for the other side.

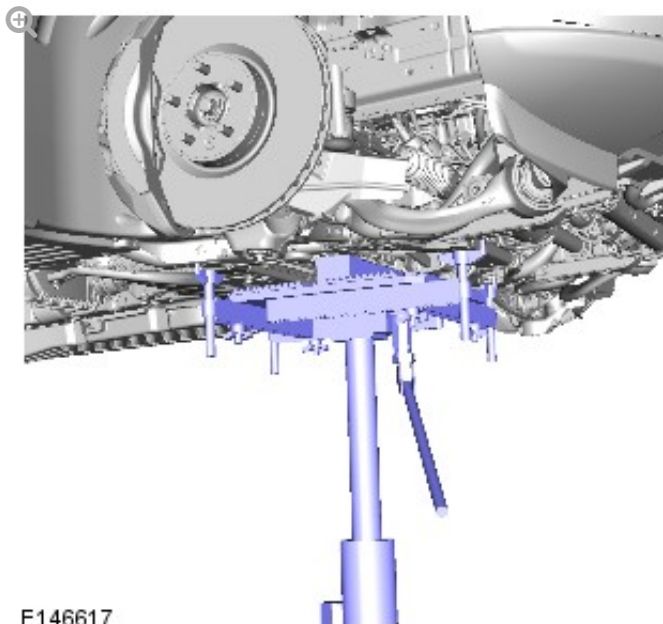


E160300



All vehicles

45.



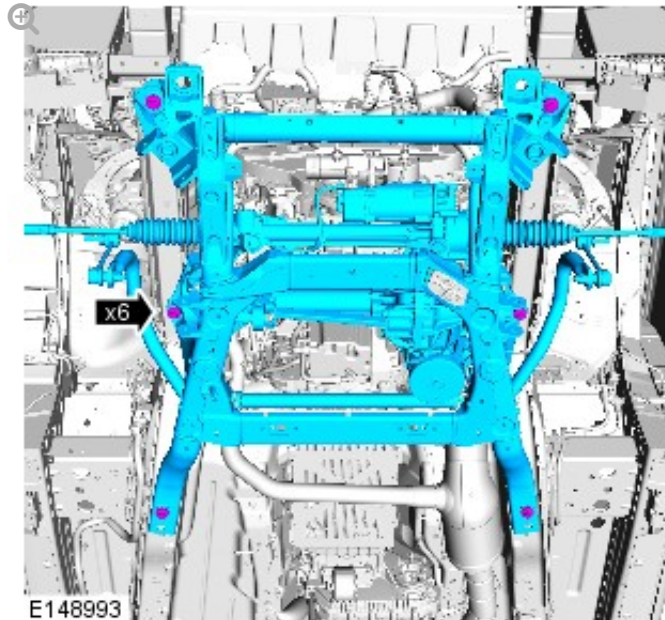
E146617

Using a suitable hydraulic jack, support the front subframe.

46.

CAUTION:

Discard the 6 bolts.

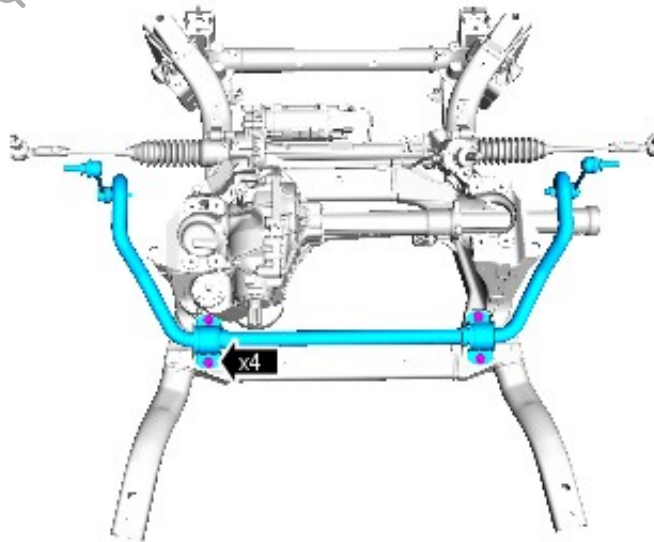


With assistance, remove the front subframe assembly.

47.

NOTE:

Do not disassemble further if the component is removed for access only.

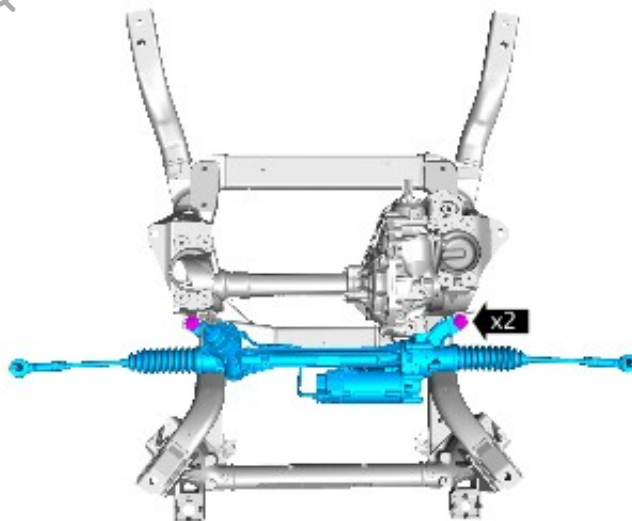


E148849

48.

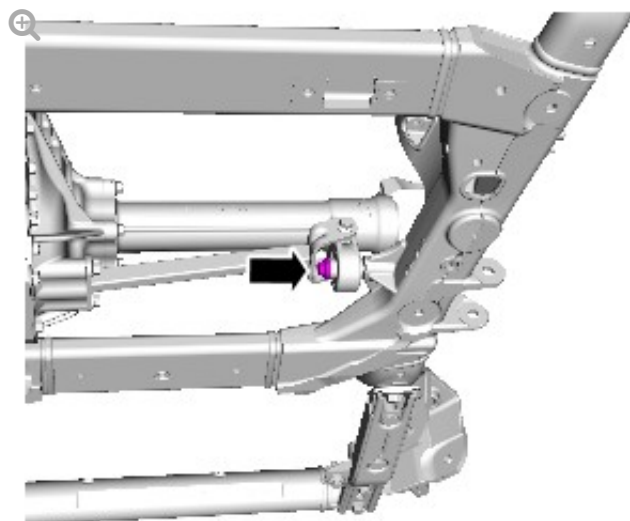
NOTE:

Discard the bolts.



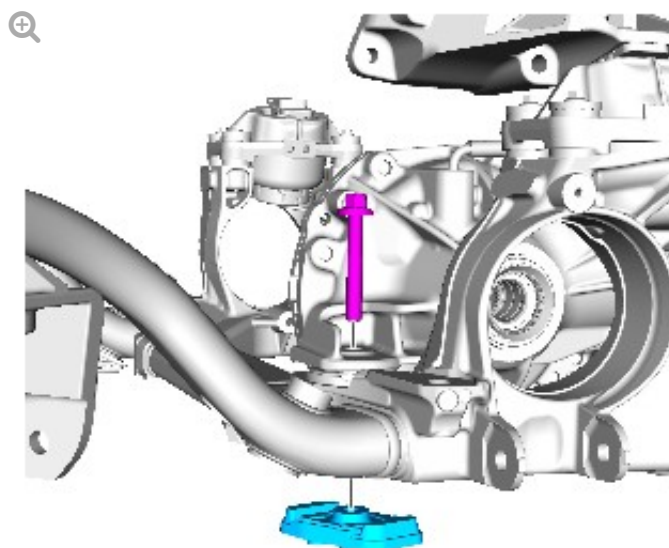
E159377

49.



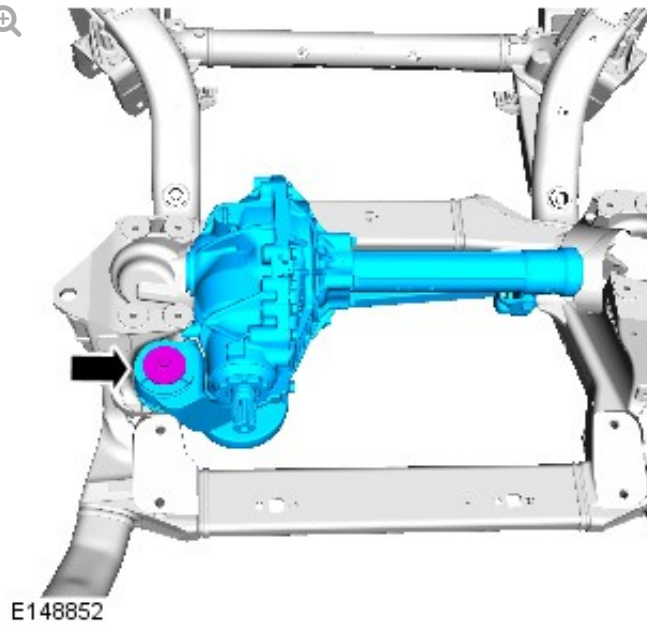
E148851

50.



E148354

51.



INSTALLATION

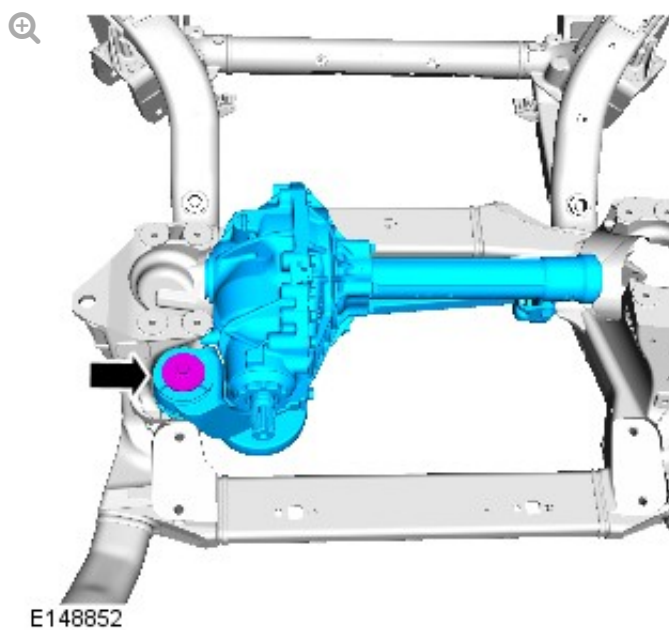


All vehicles

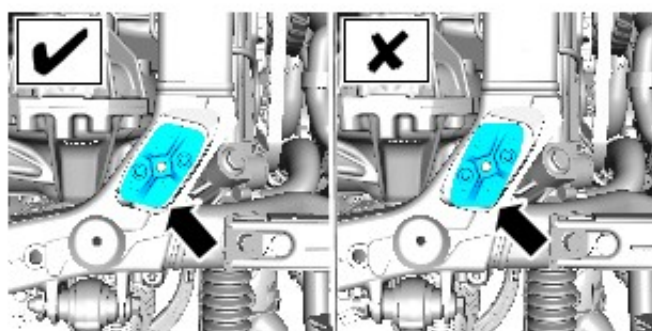
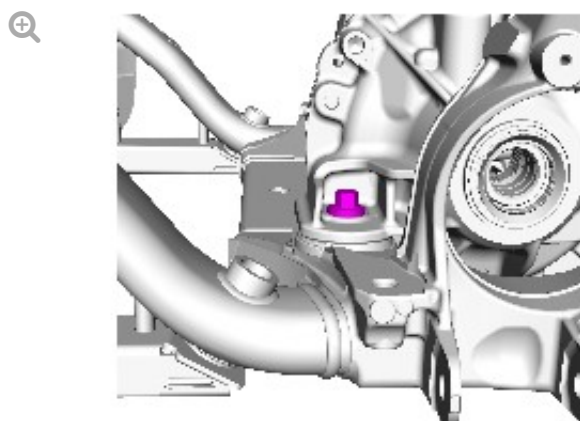
1.

CAUTION:

Install the bolt finger tight.



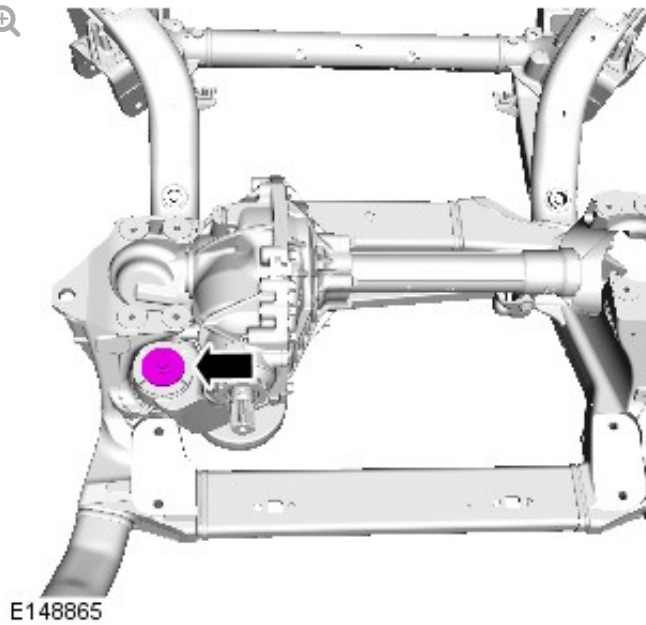
2.



E148361

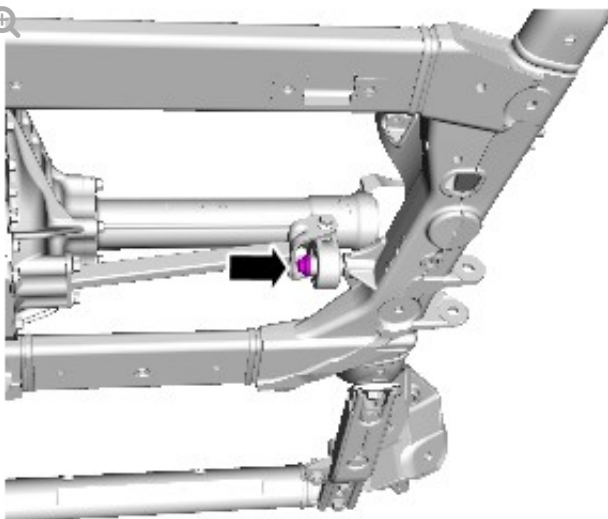
Torque: 110 Nm

3.



Torque: 60 Nm

4.

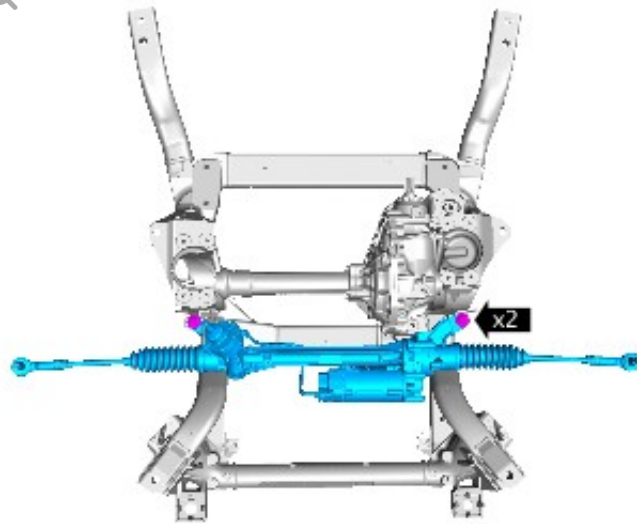


Torque: 60 Nm

5.

NOTE:

Make sure that new bolts are installed.



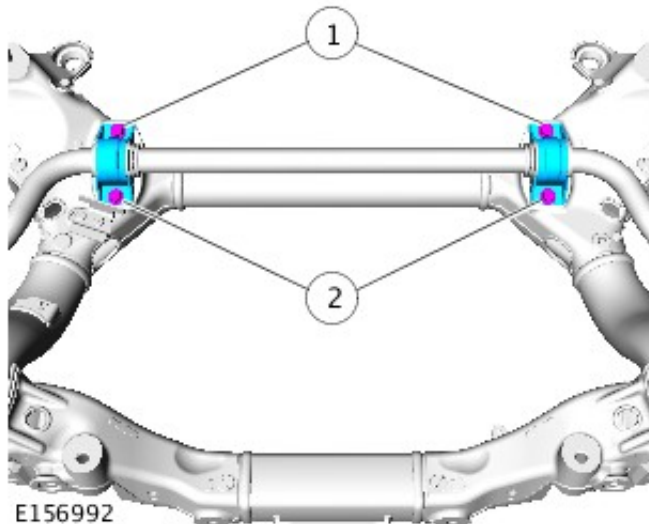
E159377

Torque:

Stage 1: **100 Nm**

Stage 2: **240°**

6.



E156992

During installation tighten the bolts in the following sequence.

Torque:

Bolts 1 **110 Nm**

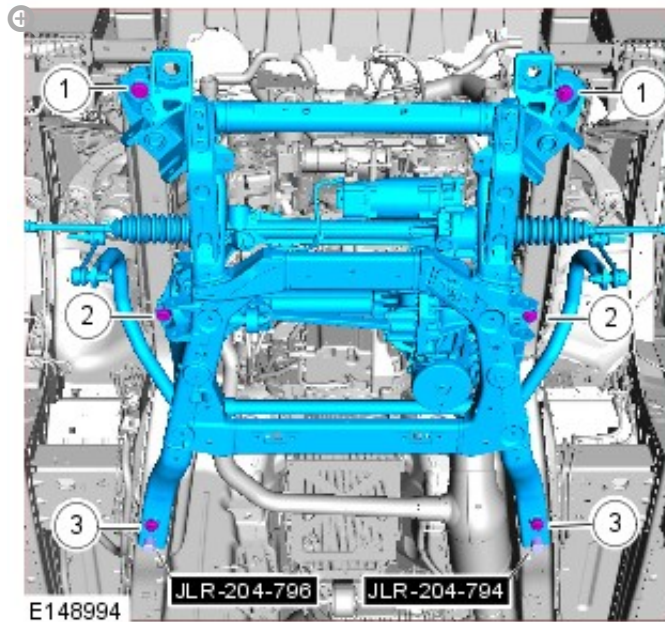
Bolts 2 **110 Nm**

Bolts 1 **110 Nm**

7.

CAUTION:

Make sure that 6 new bolts are installed.



With assistance, install the front subframe.

Special Tool(s): [JLR-204-794](#), [JLR-204-796](#)

Torque:

1

Stage 1: **100 Nm**

Stage 2: **300°**

2

Stage 1: **100 Nm**

Stage 2: **240°**

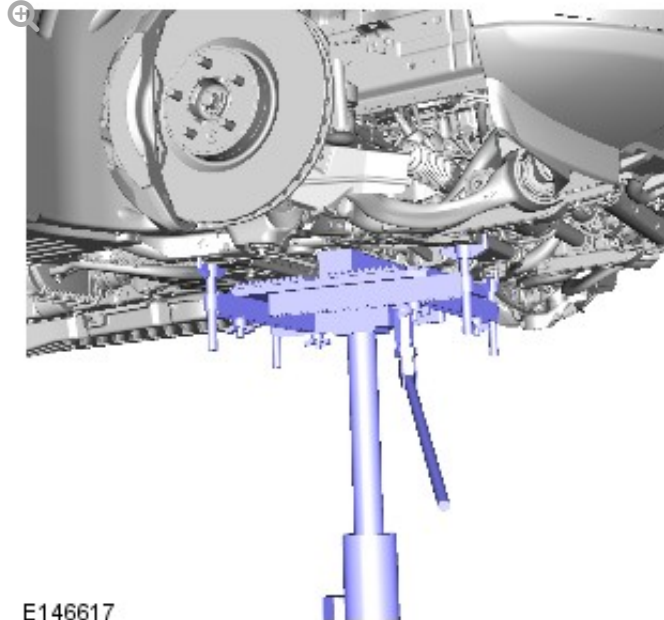
3

Stage 1: **70 Nm**

Stage 2: **120°**

8. Remove the special tools.

9.



E146617



Short wheelbase

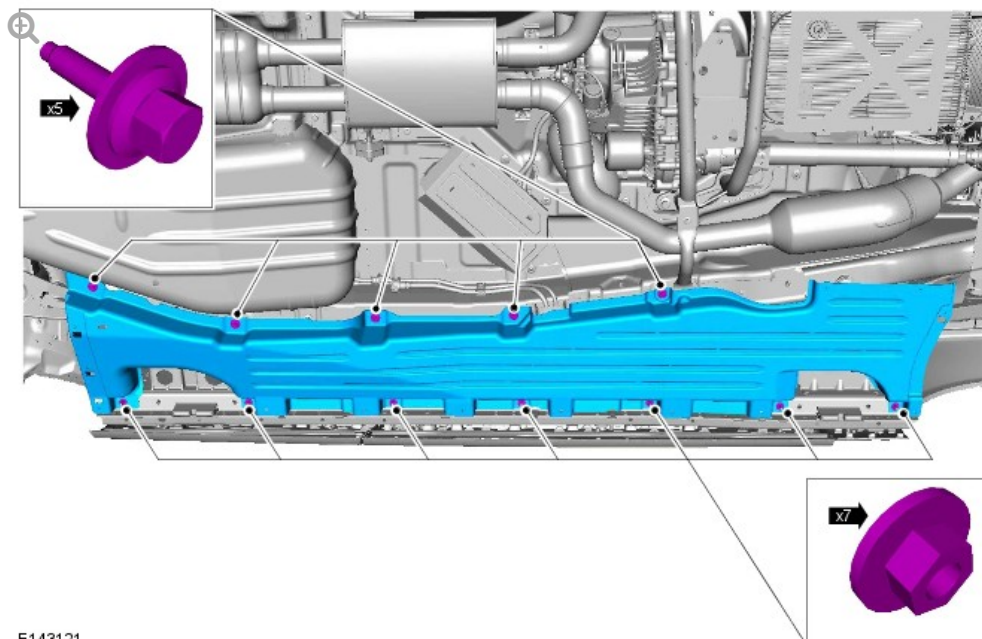
10.

CAUTION:

LH illustration shown, RH is similar.

NOTE:

Repeat the procedure for the other side.



E143121



Long wheelbase

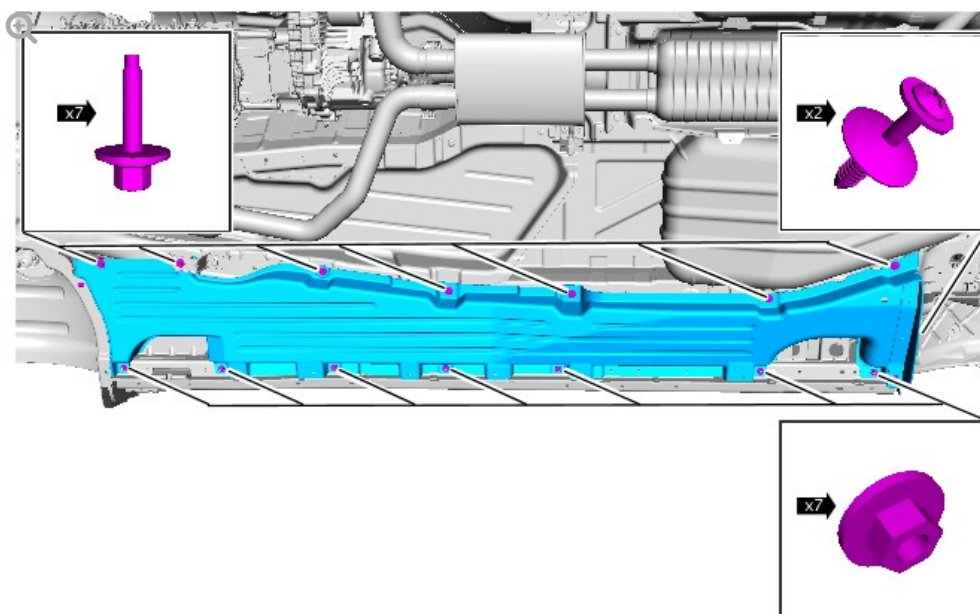
11.

CAUTION:

RH illustration shown, LH is similar.

NOTE:

Repeat the procedure for the other side.



E160300



All vehicles

12.

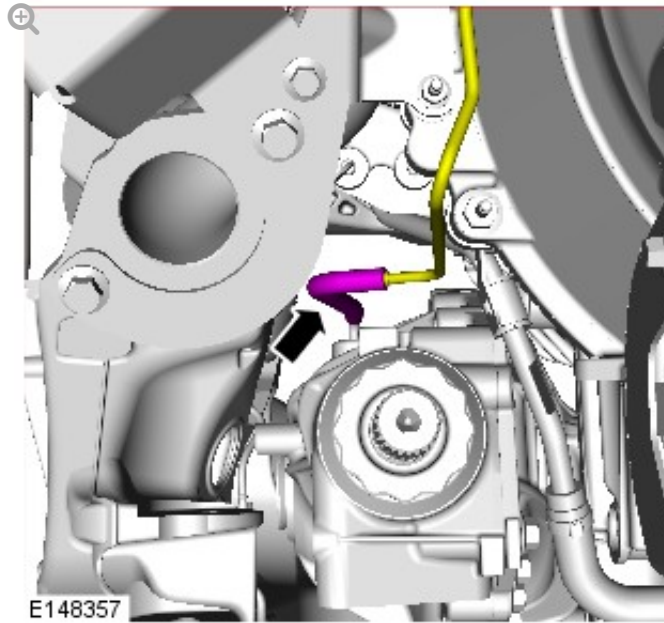
NOTE:

Repeat the procedure for the other side.

Refer to: [Rear Door Lower Moulding - Standard Wheel Base](#) (501-08 Exterior Trim and Ornamentation, Removal and Installation).

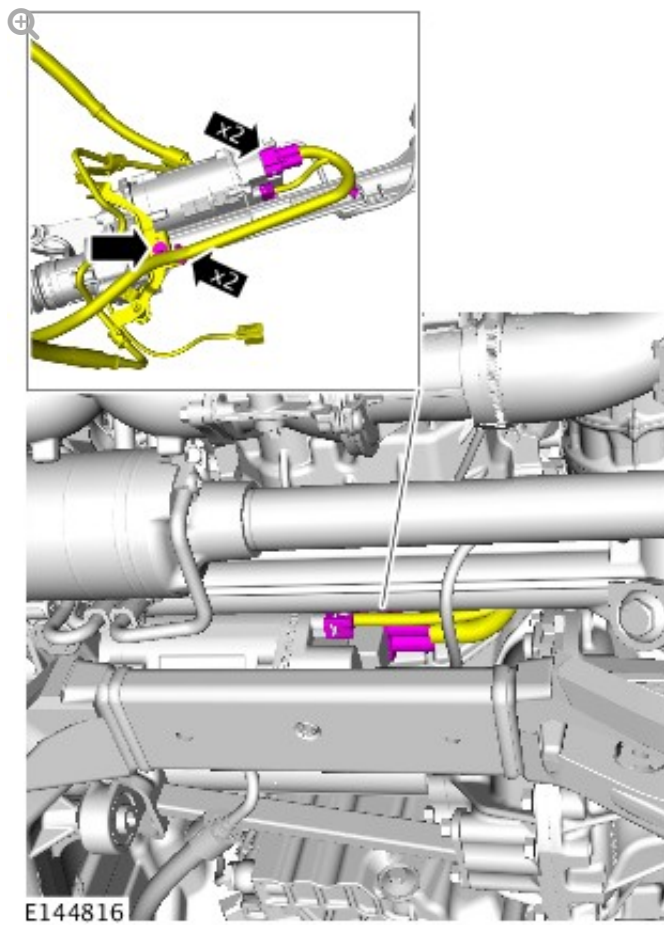
Refer to: [Rear Rocker Panel Moulding - Long Wheel Base](#) (501-08 Exterior Trim and Ornamentation, Removal and Installation).

13.



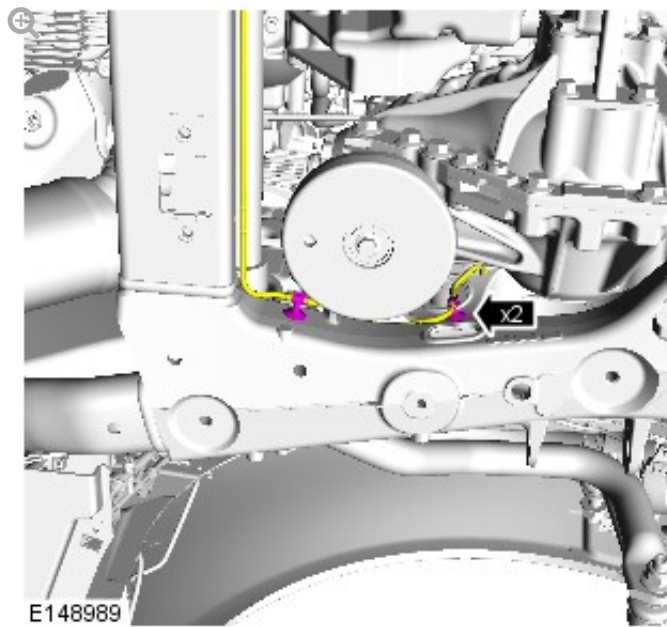
14. Refer to: [Front Driveshaft - TDV6 3.0L Diesel](#) (205-01 Driveshaft, Removal and Installation).

15.

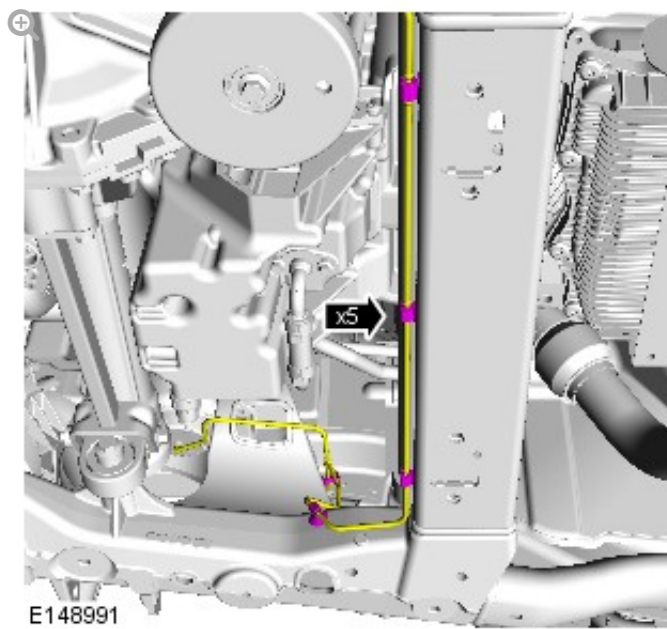


Torque: 10 Nm

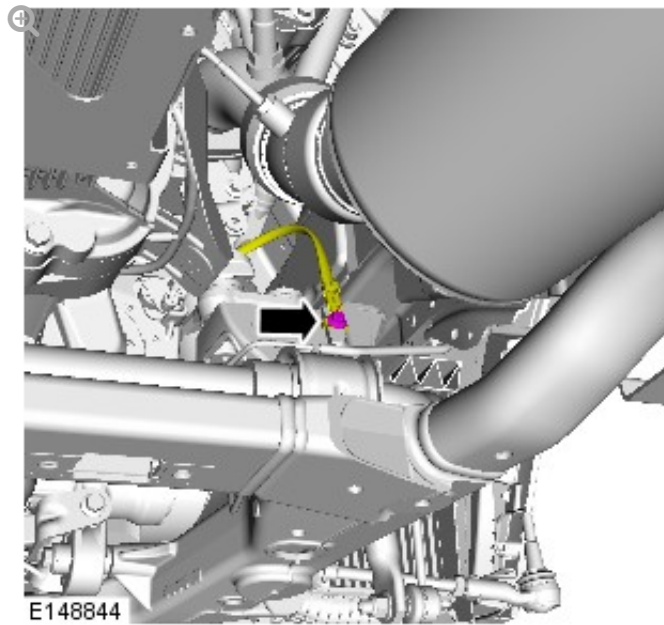
16.



17.

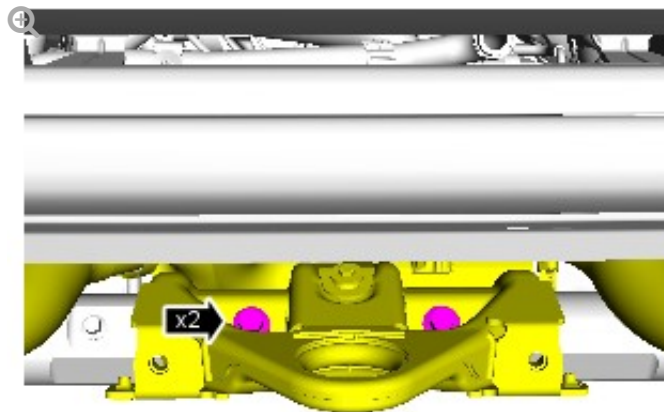


18.



Torque: 25 Nm

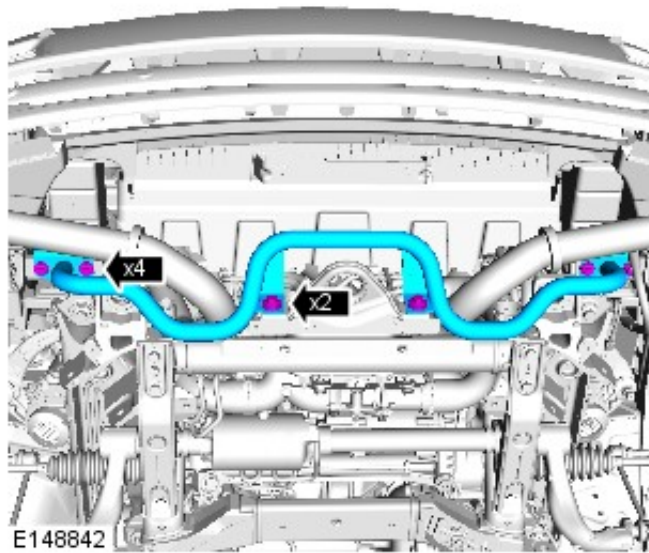
19.



E148843

Torque: 90 Nm

20.

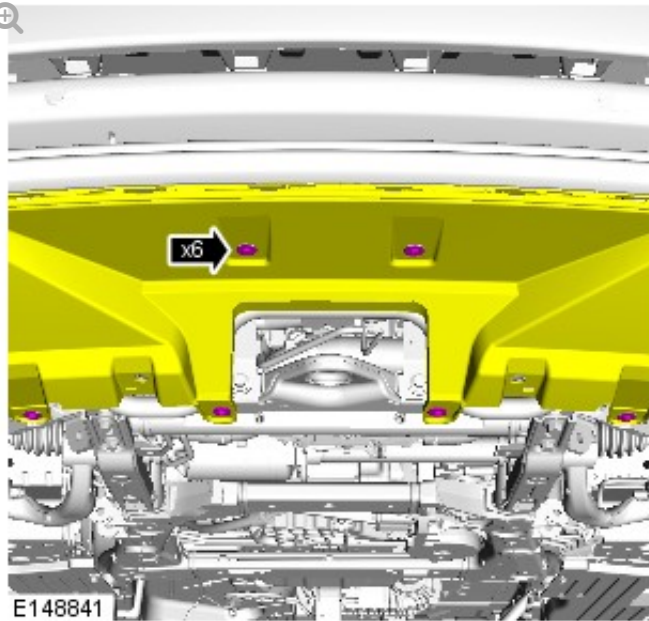


Torque:

M10 45 Nm

M12 90 Nm

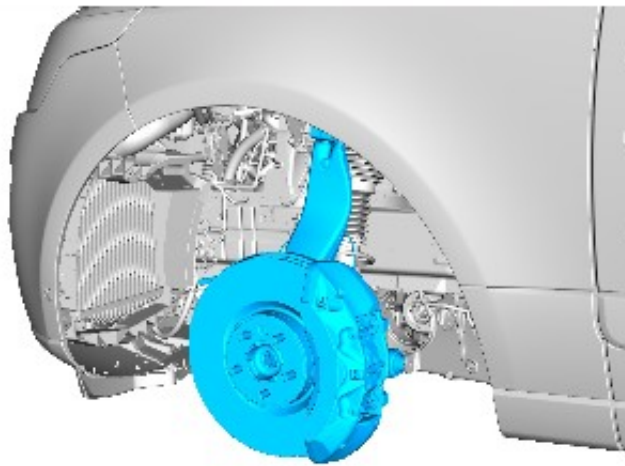
21.



22.

NOTE:

LH illustration shown, RH is similar.

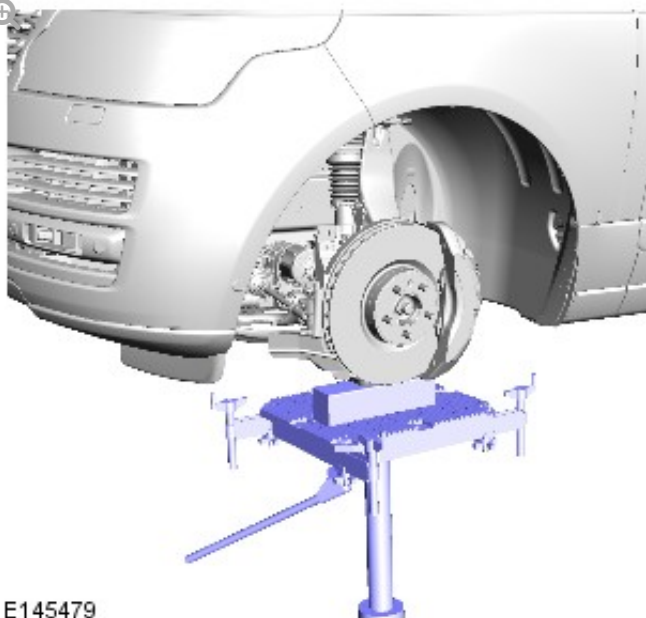


E148840

23.

NOTE:

LH illustration shown, RH is similar.

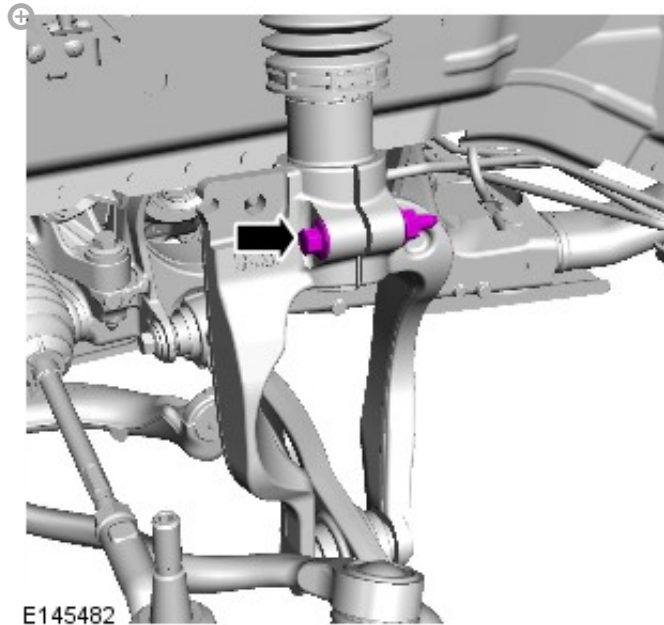


E145479

24.

NOTE:

LH illustration shown, RH is similar.



Torque: 90 Nm

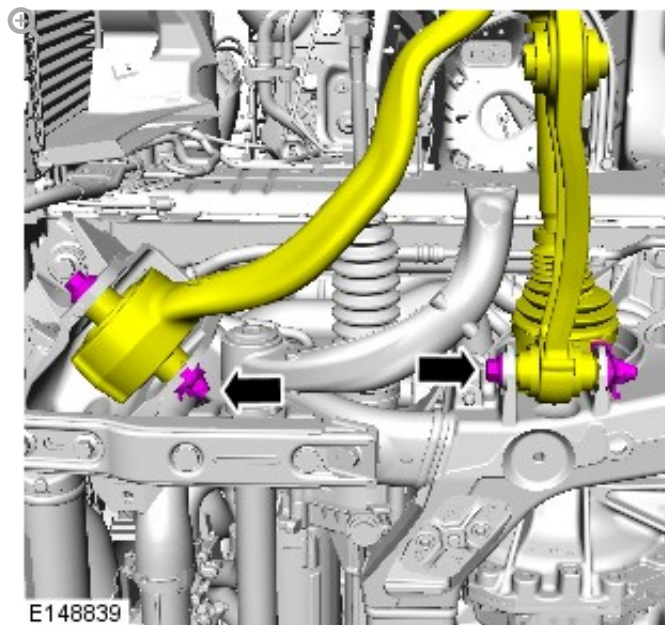
25.

CAUTION:

Only tighten the nuts and bolts when the suspension is in the normal drive position.

NOTES:

- Make sure that new nuts and bolts are installed.
- LH illustration shown, RH is similar.



Torque:

Front Lower Arm Bolt

Stage 1: **80 Nm**

Stage 2: **300°**

Rear Lower Arm Bolt

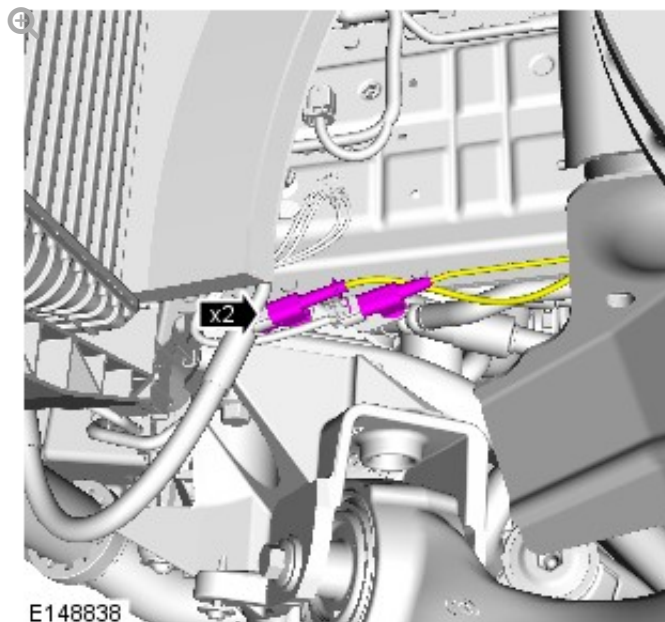
Stage 1: **100 Nm**

Stage 2: **360°**

26.

NOTE:

LH illustration shown, RH is similar.



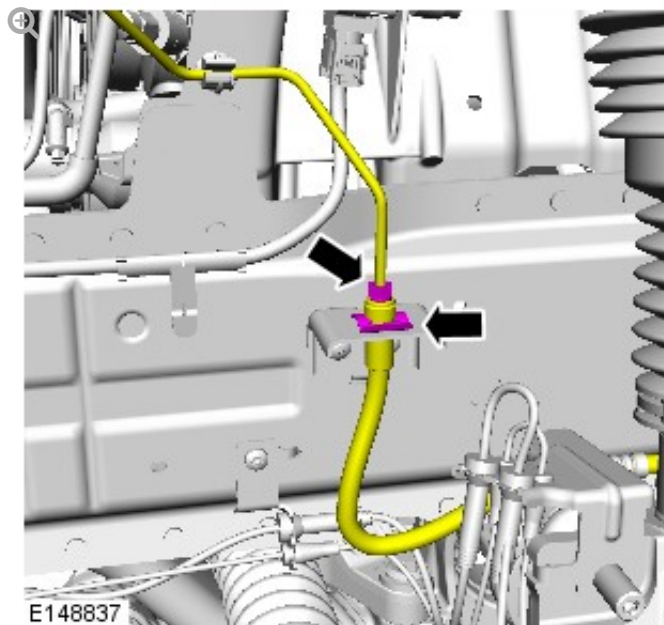
27.

CAUTION:

Make sure that the area around the component is clean and free of foreign material.

NOTE:

LH illustration shown, RH is similar.

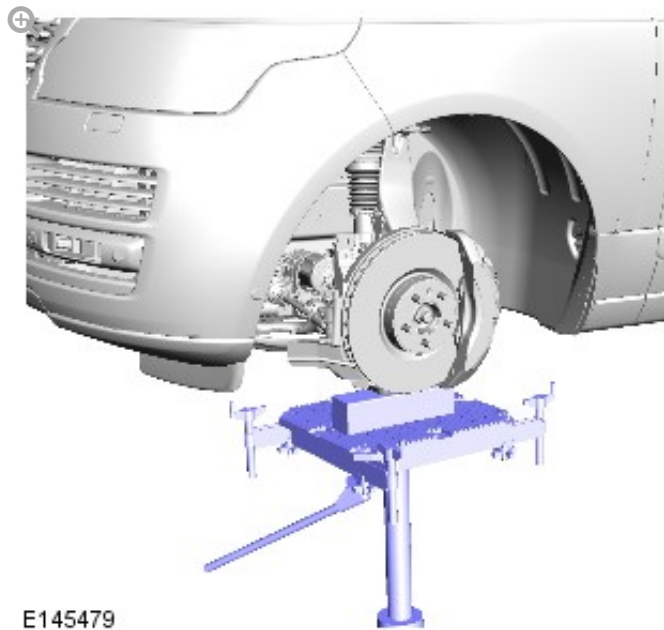


Torque: 16 Nm

28.

NOTE:

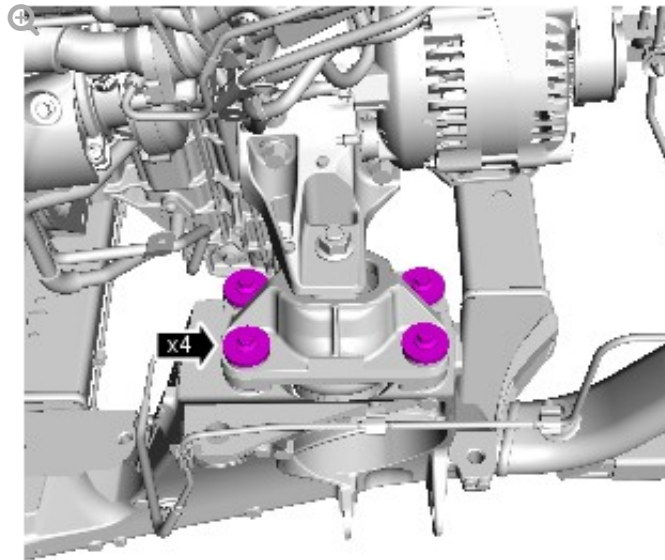
LH illustration shown, RH is similar.



29. Repeat the above 7 steps for the other side.

30. Refer to: [Front Right Halfshaft](#) (205-04 Front Drive Halfshafts, Removal and Installation).
31. Refer to: [Front Left Halfshaft](#) (205-04 Front Drive Halfshafts, Removal and Installation).

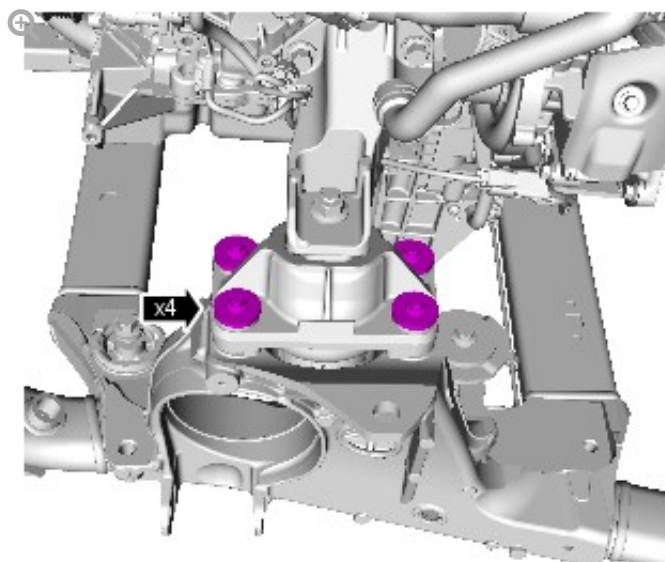
32.



E148992

Torque: 32 Nm

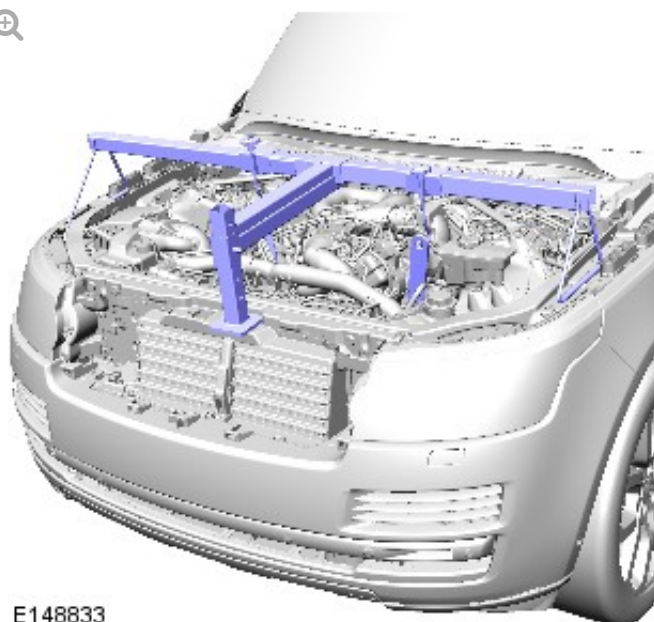
33.



E148990

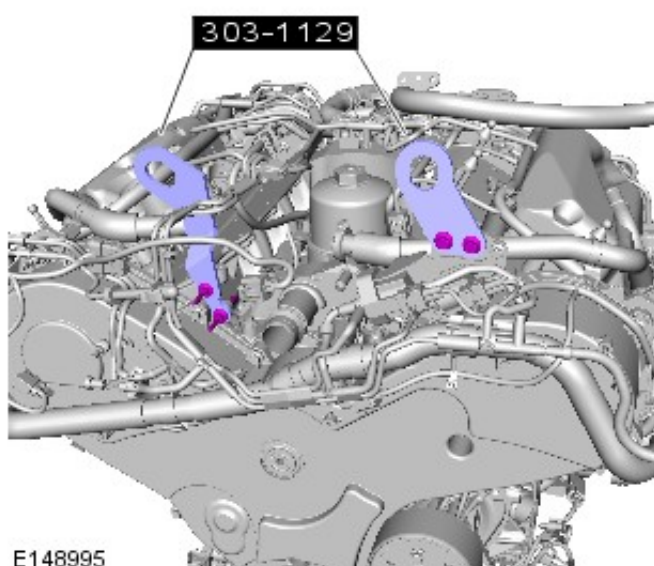
Torque: 32 Nm

34.



E148833

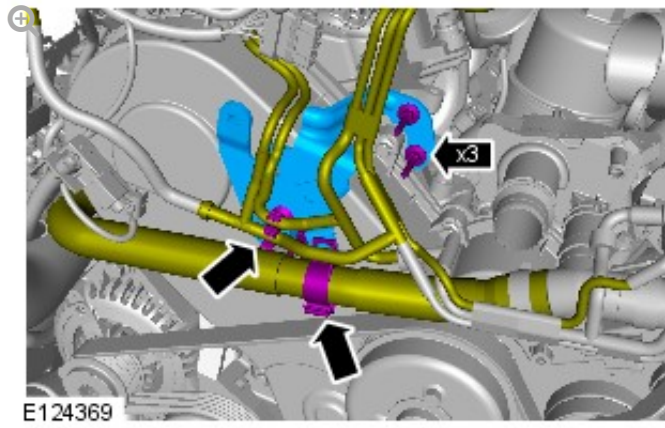
35.



E148995

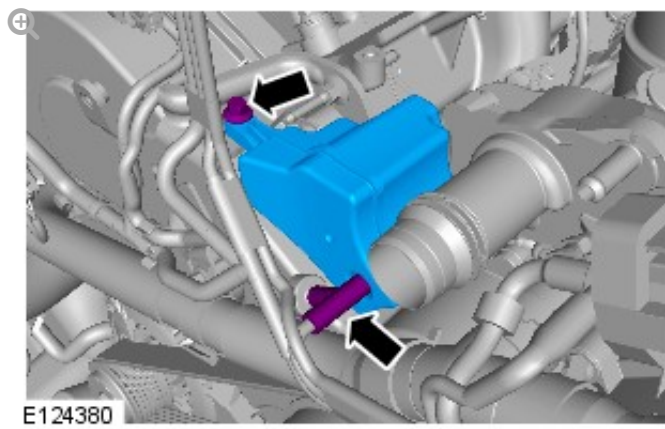
Special Tool(s): [303-1129](#)

36.



Torque: 8 Nm

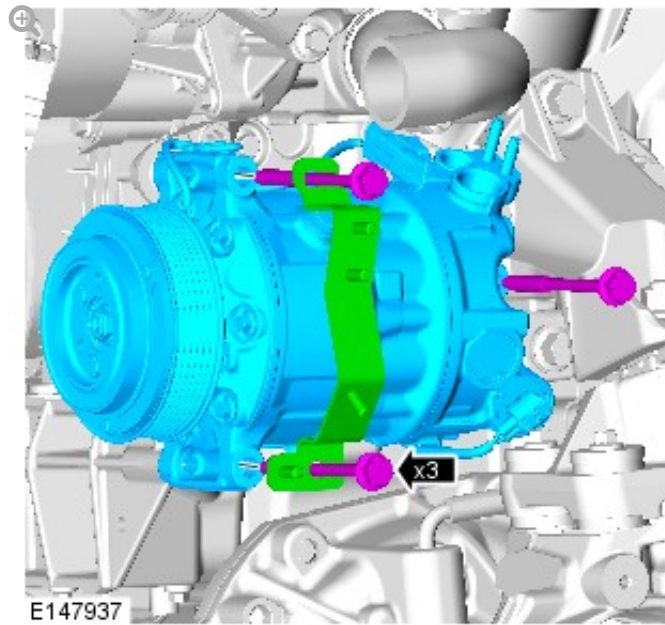
37.



Torque: 8 Nm

38. Refer to: [Intake Air Shutoff Throttle](#) (303-04A Fuel Charging and Controls - TDV6 3.0L Diesel, Removal and Installation).

39.

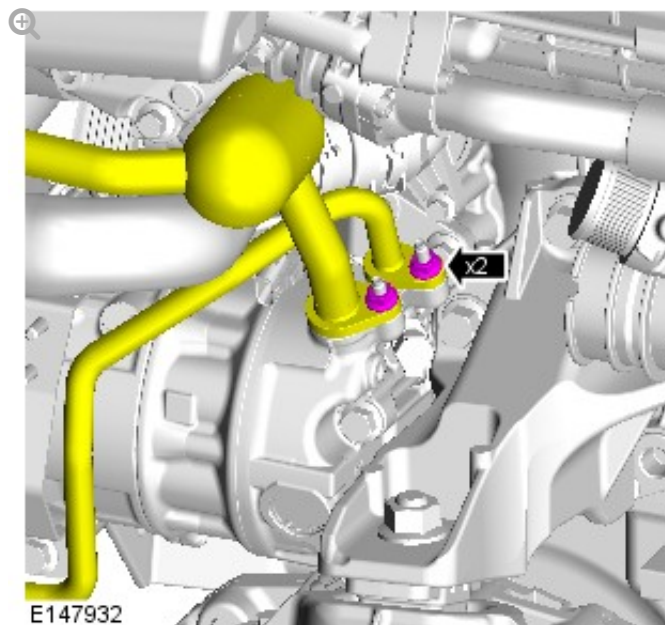


Torque: 25 Nm

40.

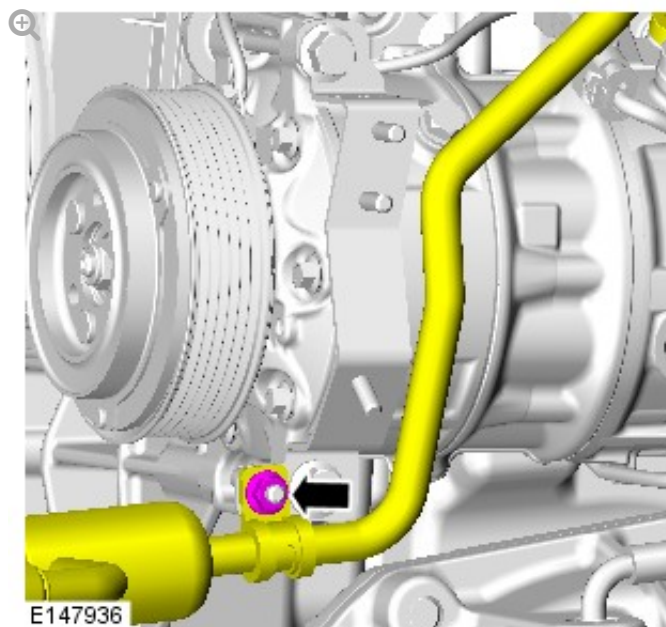
CAUTION:

A new O-ring seal is to be installed.



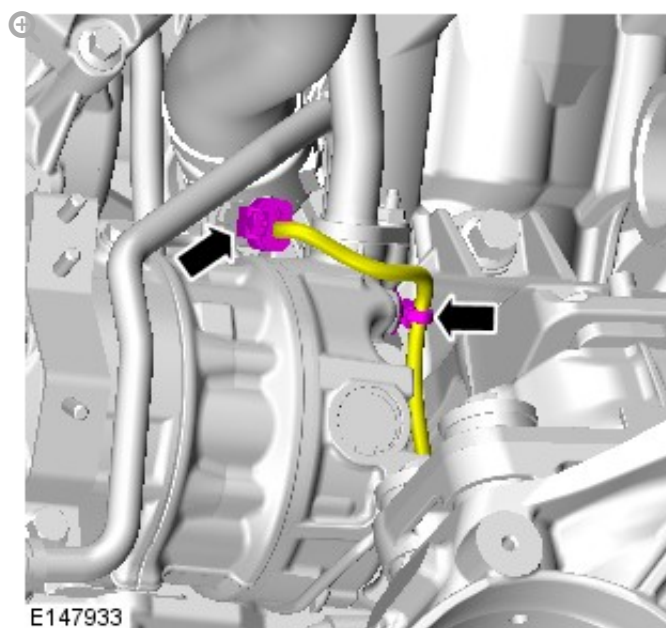
Torque: 18 Nm

41.

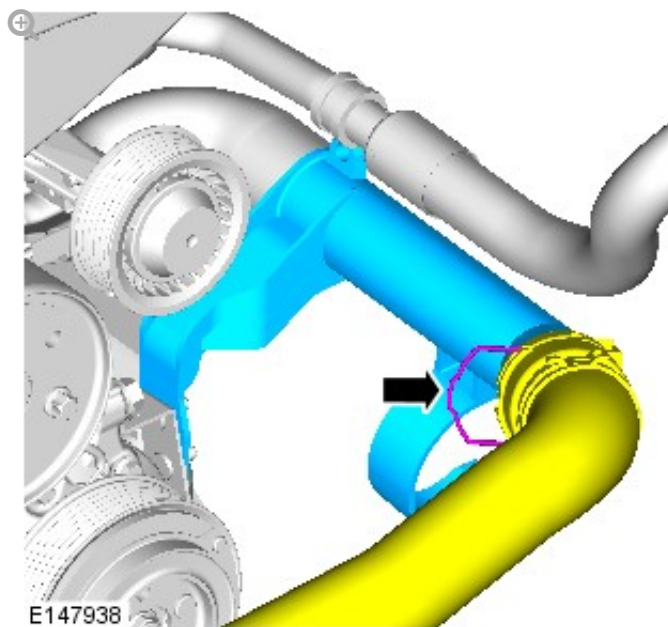


Torque: 6 Nm

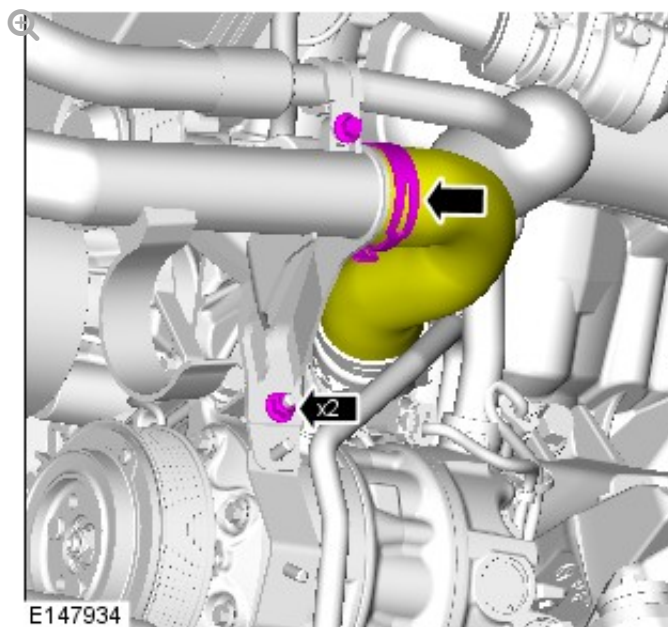
42.



43.

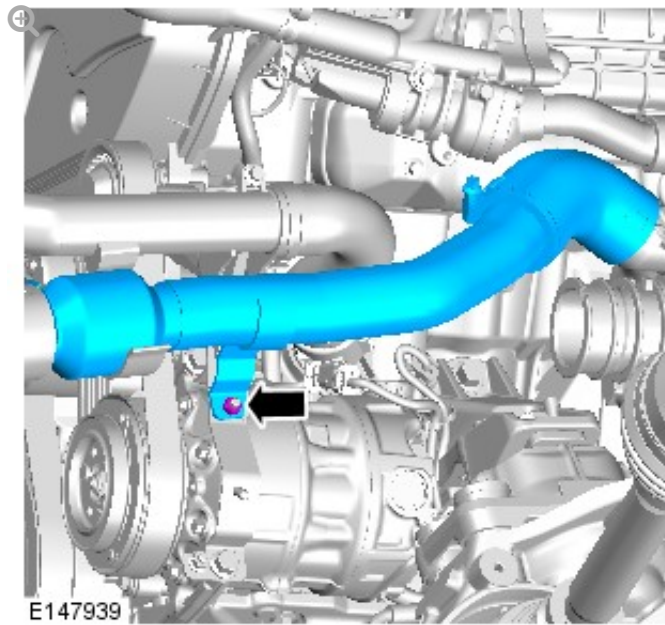


44.



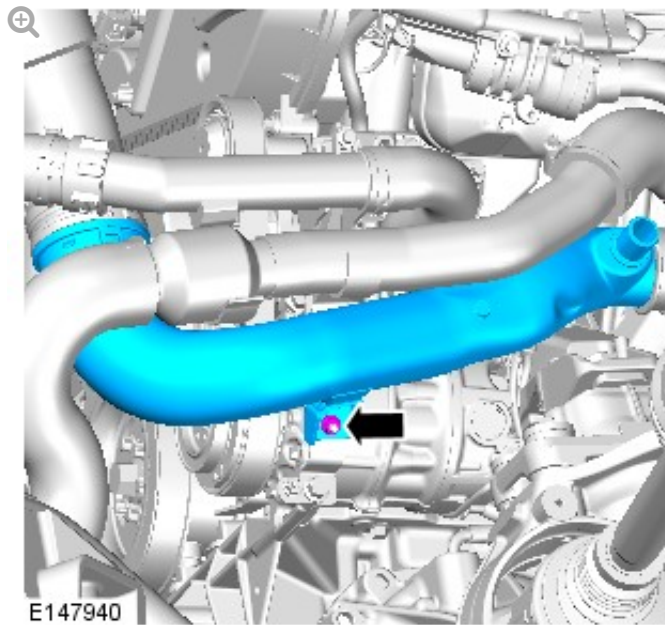
Torque: 6 Nm

45.



Torque: 6 Nm

46.

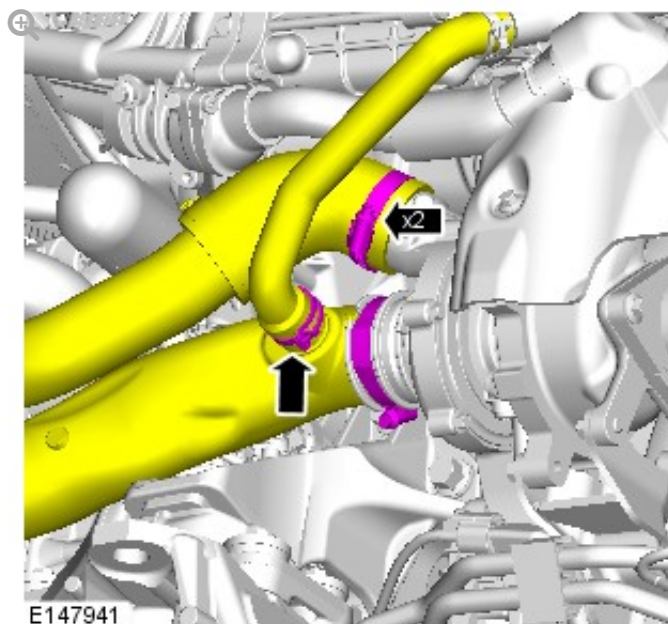


Torque: 6 Nm

47.

NOTE:

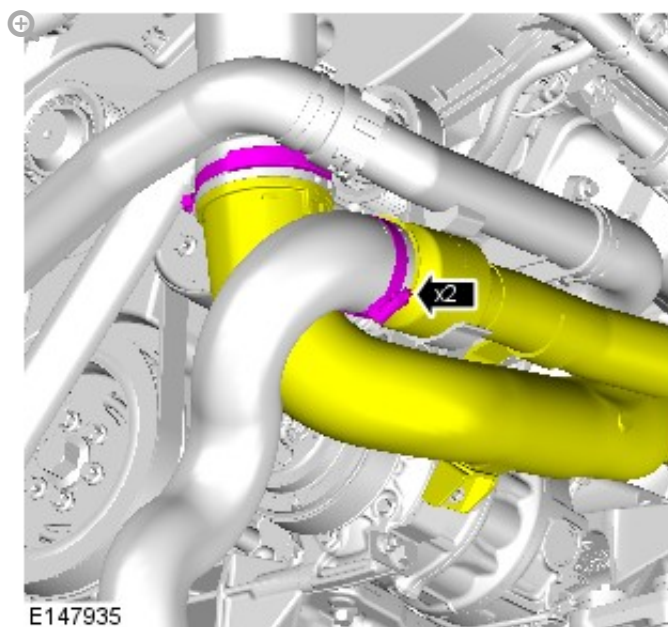
Make sure the retaining clips are correctly installed and tightened.



48.

NOTE:

Make sure the retaining clips are correctly installed and tightened.

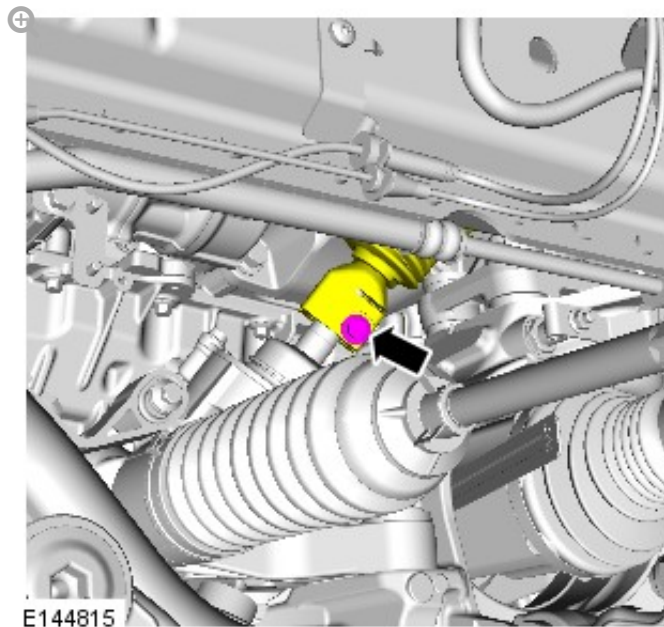


CAUTION:

Make sure that a new bolt is installed.

NOTE:

LHD illustration shown, RHD is similar.



Using a suitable tool, clean out the threads.

Torque: 30 Nm

50. Refer to: [Generator](#) (414-02A Generator and Regulator - TDV6 3.0L Diesel, Removal and Installation).
51. Refer to: [Radiator](#) (303-03A Engine Cooling - TDV6 3.0L Diesel - Gen 2/TDV6 3.0L Diesel, Removal and Installation).
52. Connect the battery ground cable.
Refer to: [Specifications](#) (414-00 Battery and Charging System - General Information, Specifications).

53. Refer to: [Brake System Bleeding - Vehicles With: Standard Brakes](#) (206-00 Brake System - General Information, General Procedures).
Refer to: [Brake System Bleeding - Vehicles With: High Performance Brakes](#) (206-00 Brake System - General Information, General Procedures).
54. Refer to: [Four Wheel Alignment](#) (204-00 Suspension System - General Information, General Procedures).